AGC Data Book

AGC Inc.

March 2026

Your Dreams, Our Challenge

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Corporate Information

Company Profile

Company Name	AGC Inc.
Head Office	〒100-8405 Shin-Marunouchi Bldg., 1-5-1Marunouchi, Chiyoda-ku, Tokyo, JAPAN
Incorporated	June 1, 1950〔Founded : September 8, 1907〕
Representative	Representative Director, President& CEO Yoshinori Hirai
Book closing date	December 31
Capital	90,873 million yen
Consolidated Revenue	20,588 billion yen
Subsidiaries	Subsidiaries:192 including 155 companies overseas (Consolidated subsidiaries 175 including 140 companies overseas) Affiliate companies: 26 including 16 companies overseas
Employees Consolidated	52,896

Stock Information

Stock Code	5201
Stock Exchange Listings	Tokyo
Minimum Trading Units	100
Number of shares issued	217,434,681
Stock Information	View the stock information page
Introduction of Executives	View the Officer information page

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Personal Profile



Representative Director,
President& CEO

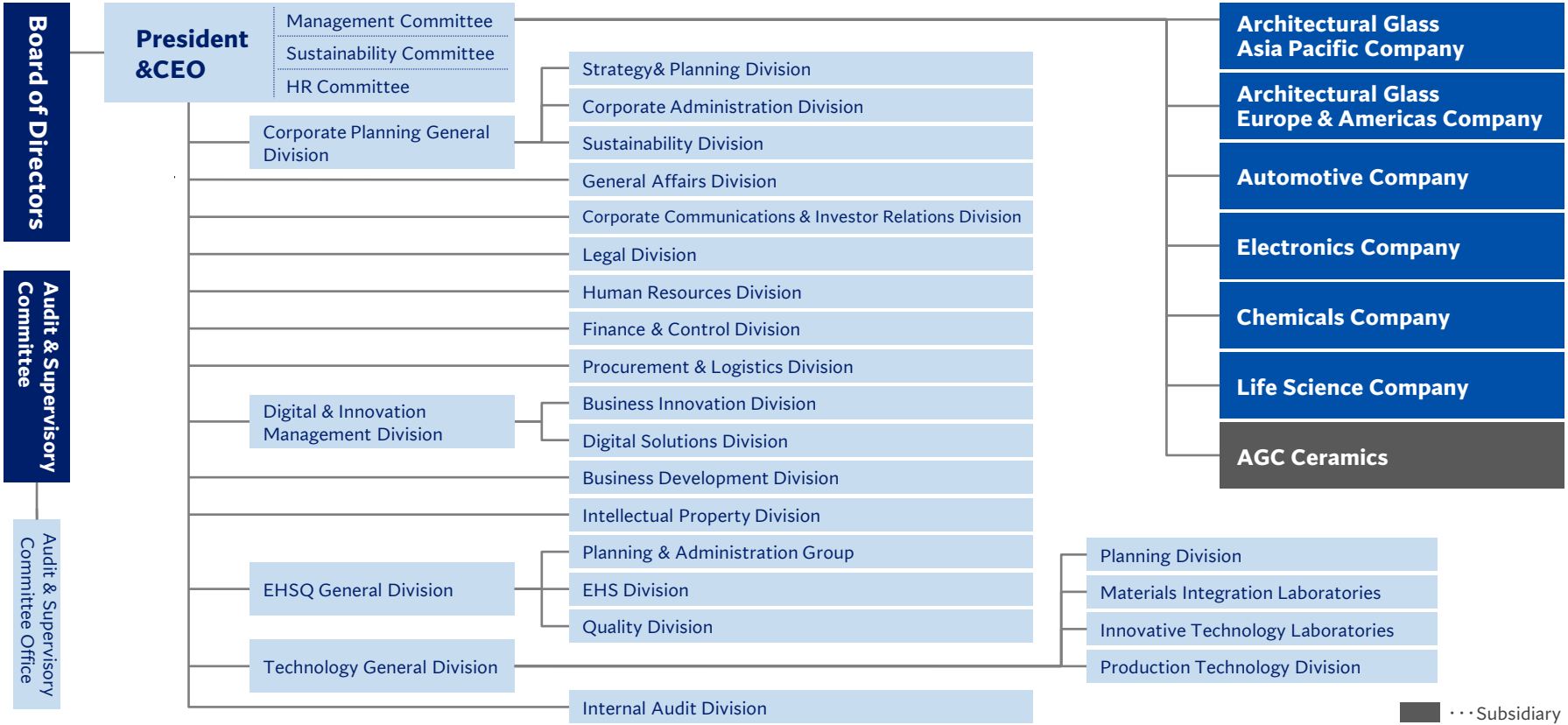
Yoshinori Hirai, Ph.D.

Education/ Business Experience

1978	Graduated from Fukui Prefectural Fujishima High School
1982	Graduated from the University of Tokyo
1987	Ph.D. on Physics from the University of Tokyo
1987	Joined Asahi Glass Co., Ltd. (current AGC Inc.)
1997	LCD Device Group Leader R&D Center for Electronic Technology, Electronics Business Division
2005	Member of the Board and GM, OPTREX
2008	Senior Executive Vice President, OPTREX
2009	GM of Business Planning Office Electronics Company (current AGC Inc.)
2011	GM of Business Development Office
2012	Executive Officer and GM of Business Development Office
2014	Member of the Board Senior Executive Officer GM of Technology General Division
2016	Member of the Board Senior Executive Officer CTO and GM of Technology General Division
2018	Representative Director Executive Vice President and CTO
2021	Representative Director President& CEO

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AGC Group Organization Chart



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AGC's R&D center & Plants in Japan

Name	Address	Product
Kansai Plant (Amagasaki factory)	〒660-0857 2 Nishimukaijima-cho, Amagasaki-shi, Hyogo Pre.	FPD glass substrates
Kansai Plant (Takasago factory)	〒676-8655 5-6-1 Umei Takasago-shi Hyogo Pre.	FPD glass substrates, Semiconductor production
AGC Yokohama Technical Center (former Research Center/Keihin Plant)	〒230-0045 1-1 Suehiro-cho, Tsurumi-ku, Yokohama-shi, Kanagawa Pre.	Research & Development Figured glass, Automotive glass
Chiba Plant	〒290-8566 10 Goikaigan, Ichihara-shi, Chiba Pre.	Chlor-alkali, Fluoropolymer, Biopharmaceutical CMO service
Aichi Plant	〒470-2394 1 Asahi, Taketoyo-cho, Chita-gun, Aichi Pre.	Automotive glass and Glass for Solar Cells
Kashima Plant	〒314-0195 25 Higashiwada, Kamisu-shi, Ibaraki Pre.	Flat glass, Fluoropolymer, Sodium bicarbonate
Sagami Plant	〒243-0301 426-1 Sumida, Aikawamachi, Aiko-gun, Kanagawa Pre.	Automotive glass

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AGC Group's Global Network

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The Group operates in approximately 30 countries and regions in the world.

Total 192 subsidiaries (37 in Japan/155 overseas)

Europe

93

Major subsidiaries	
AGC Glass Europe	Belgium
AGC Automotive Europe	Belgium
AGC Flat Glass Czech a.s.	Czech Republic
AGC Biologics A/S	Denmark

Japan/Asia

88

Major subsidiaries			
AGC Ceramics Co., Ltd.	Japan	AGC Automotive (Suzhou) Inc.	China
AGC Glass Products Co., Ltd.	Japan	AGC Display Glass (Huizhou) Co., Ltd..	China
AGC Glass Kenzai Co., Ltd.	Japan	AGC Fine Techno Korea Co., Ltd.	Korea
Ise Chemicals Corporation	Japan	P.T. Asahimas Chemical	Indonesia
AGC Display Glass Taiwan Co., Ltd.	Taiwan	Vinythai Public Company Limited	Thailand

Americas

11

Major subsidiaries	
AGC Flat Glass North America, Inc.	The United States
AGC Biologics Inc.	The United States
AGC America, Inc.	The United States
AGC Capital, Inc.	The United States

As of December 31, 2025

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Products with Top Market Share

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	Architectural Glass	Automotive	Electronics	Chemicals	Life Science
Core Businesses	 Float glass **Top share** * in Europe, Japan, etc	 Automotive glass **Top share** * Worldwide	 Glass substrates for TFT-LCD/OLED **No.2** Worldwide *	 Caustic soda and PVC ◀ Mountain of salt used as a raw material **No.1** in Southeast Asia *	
Strategic Businesses		 Car-mounted cover glass **Top share** * Worldwide	 EUV lithography photomask blanks **No.2** Worldwide *	 Fluorinated resins (Fluon® ETFE) **No.1** Worldwide *	 CDMO services for Ex vivo gene therapy products **No.1** Worldwide *

*Based on fiscal 2026 estimates by AGC

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AGC Group's Glass Variation

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- As a comprehensive glass manufacturer, the AGC Group manufactures a wide variety of glass products.

Soda-lime glass

- Building
- Automotive
- TN/STN

Alkali-free glass

- TFT-LCD

Aluminosilicate glass

- Chemical Strengthening glass

Quarts

- Synthetic quartz

Borosilicate glass

- Heat-resistant cookware
- Physics and chemistry equipment

Bismuth-based glass

- Optics use

Glass Manufacturing Bases (Float Lines)

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- Total: 26 float lines (two lines are managed by equity method affiliate)

Europe

7 plants | 11 floats

City	Country / State	Number of FLs
Moustier	BEL	3
Mol	BEL	1
Seingbouse	FRA	1
Osterweddingen	DEU	1
Cuneo	ITA	1
Sagunto	ESP	1
Retenice	CZE	3

Japan / Asia Pacific

9 plants | 12 floats

City	Country / State	Number of FLs
Kashima	JPN	1
Aichi	JPN	2
Suzhou	CHN	1
Cikampek	IDN	2
Surabaya	IDN	2
Samut Prakarn	THA	1
Chon Buri	THA	1
Taloja (*)	IND	1
Roorkee (*)	IND	1

Americas

1 plant | 1 float

City	Country / State	Number of FLs
Richmond	U.S.A / KY	1

South Americas

1 plant | 2 floats

City	Country / State	Number of FLs
Guaratingueta	Brazil / Sao Paulo	2

As of February 2023

(*) Equity method affiliate

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Architectural Glass

Architectural Glass Products

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General name	Representative product name	Product description
Float glass	Float glass	Transparent flat sheet glass. The most common type of flat glass used for construction purposes. Thickness :from 3mm to 19mm. Maximum size: around 3m in width, around 10m in height (Maximum size vary depending on area and thickness).
Laminated glass	Lamisafe™/Lamisafe Security™ Lamisafe Shelter™	Consists of two pieces of glass with an interlayer film inserted in between. Very few fragments scatter even when glass breaks and has excellent penetration resistance. Excellent for disaster/crime prevention. Also has UV blocking properties.
IGU (Insulated Glass Unit)	Pairglass™ MyMute™ （noise reduction）	Consists of two pieces of glass with dried air and heat-insulating gas enclosed in between. Provides excellent heat insulation, energy conservation and dew condensation prevention.
Low-E IGU	Sunbalance™ Pairplus™ （renovations）	IGU with special metal film coating on the inner surface of one of the glass. Provides enhanced heat insulating/shielding performance.
Tempered glass	Temperlite™/School Temper™ Mistron Ace™/Home Mistron	Safety glass with the strength 3~5 times greater than ordinary glass, produced through special heating and cooling processes. Superior resistance to breakage, and even when breaks, it fragments into small pebble-sized pieces.
Tempered fire resistance glass	Myboka™	Fire-resistant glass with heat-resisting properties, produced through a special tempering process. This non-wired glass provides a clear view.
Mirror	Sunmirror™G	Environmentally friendly mirror with lead-free back paint coating.
Colored glass	LACOBEL™/MATELUX™	Painted glass produced by adding special paint coating on the back surface of glass. Mainly used as interior glass for commercial facilities.
Figured glass	Kasumi™	Flat glass with a template pattern on one side. Provides privacy while letting light through. Also brings interior effect .
Wired glass	Hishiwire™ /Crosswire	Glass with wire mesh enclosed. Used for fire protection doors.

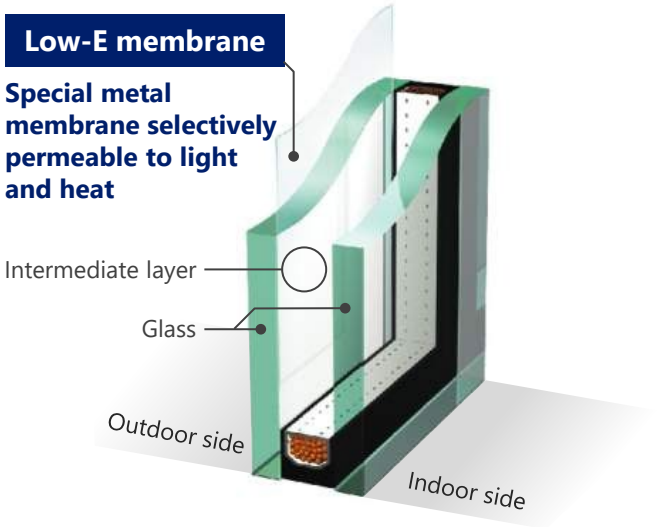
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Low-E double glazing glass

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- Selectively transmit light and heat and helps to improve the energy efficiency of buildings
- **Approximately 78% less heat transfer than single pane glass***

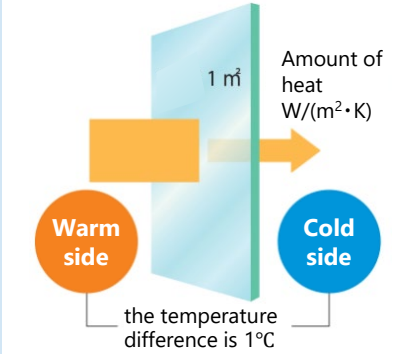
Structure of Low-E double glazing glass



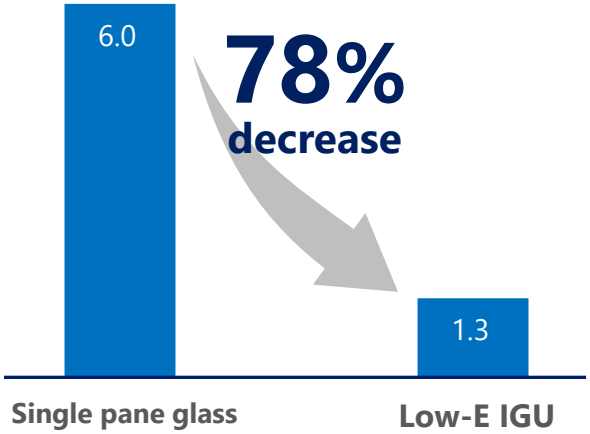
Effect of Low-E double glazing glass

【Thermal transmission rate】

Amount of heat passing per 1 ㎡ for 1 hour with a temperature difference between inside and outside is 1 ℃.



Change in thermal transmittance with the use of Low-E*



* Comparison of thermal transmittance between 3 mm float glass and gas-filled Low-E IGU

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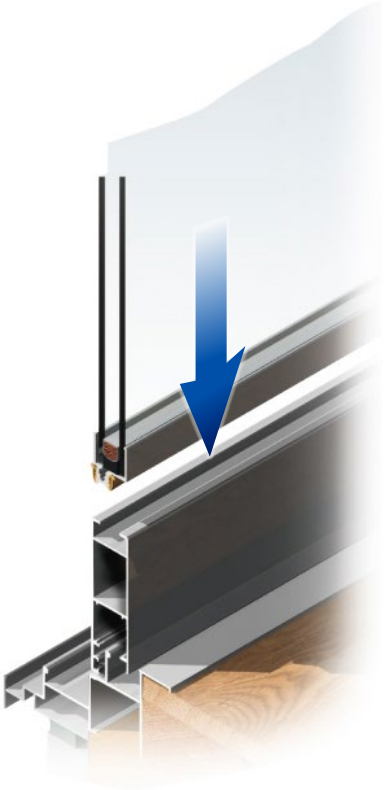
Energy-saving Home-improvement Products

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PairPlus™

- Heat insulation
- Solar control
- Anti condensation

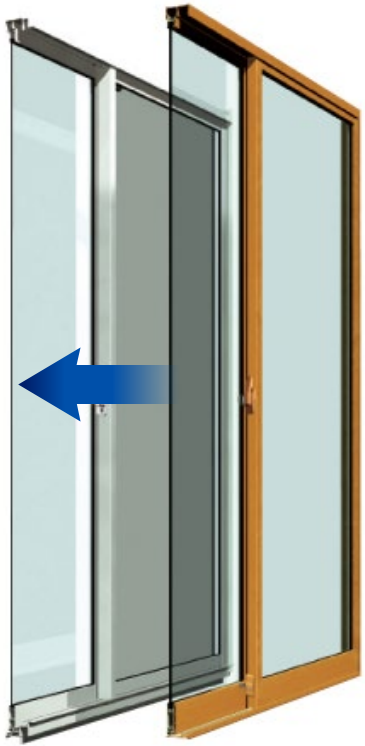
This product allows single-pane glass to be replaced with Low-E double-glazing glass without changing the existing sash.



Mado2™

- Heat insulation
- Solar control
- Anti condensation
- Noise reduction

Additional layer of glass added to the existing glass/window sash to enhance functionality of the window.



Energy-saving Products for Building renovation

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ATTOCH™

- Heat insulation
- Solar control
- Anti condensation

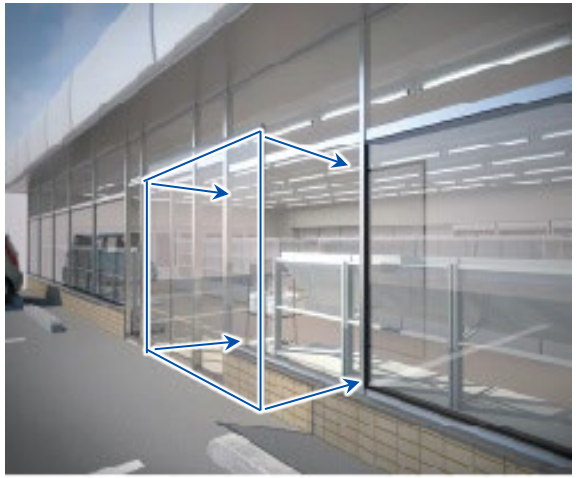


ATTOCH™ converts single-pane fixed windows of office buildings and stores into energy-saving insulated glass units simply by installing low-E glass from the inside.

Product Lines

WILD ATTOCH™

WILD ATTOCH is enables single-story retail shops to do energy-saving reform without disrupting store operations.

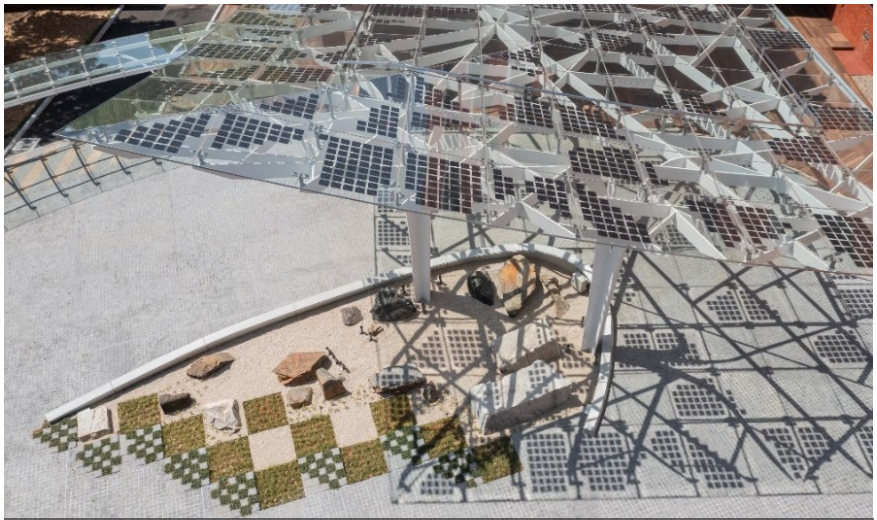


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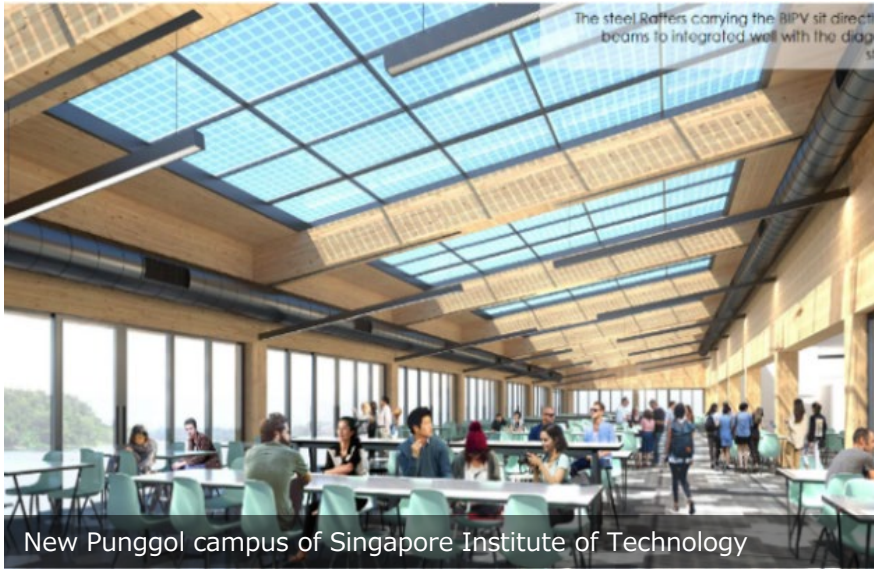
Photovoltaics-embedded glass: SunEwat™

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- Glass that can generate electricity by solar power
- Encapsulates a solar power generation cell in two glass plates
- Realizes both energy creation and design performances and help bring about a carbon-neutral society



Entrance canopy of Global Zero Emission Research Center of AIST



New Punggol campus of Singapore Institute of Technology

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Display-integrated mirror: Mirroria™

- With AGC's unique optical design technology, both "high reflectivity" and "display visibility" are achieved.
- By combining various software, cameras, and sensors, a wide range of services can be offered.



Enhancing customers experience and improving salon operations at Takara Belmont Co.,





Digital signage advertising in office restrooms in Shin-Marunouchi Building. (demonstration experiment)

* Building Integrated Photovoltaics

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Safety Glass/Fire-resistant Glass

Lamisafe Shelter™

(Laminated glass)

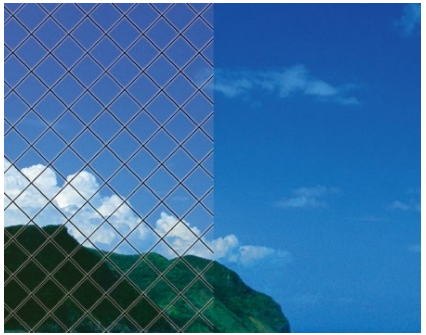
This is a "Disaster Prevention Safety Laminated Glass " with a flexible and tough special film of 60mil (about 1.5mm) or more in thickness sandwiched between two sheets of glass.

It has excellent penetration resistance, and the strong adhesion between the special film and glass makes the glass shatterproof.



Fire-resistant Glass

AGC's fire-resistant product lineup consists of wired glass and non-wired heat-resistant tempered glass "Myboka™."



Features of Non-wired tempered fire resistance glass

1	2	3
Provides a clear view	**No thermal breakage** (as compared to wired glass)	**Lighter-weight & easier open/close than wired glass**

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Decorative Glass

Lacobel™

Colored glass for interior use

Lacobel is clear float glass featuring a high-quality paint on one side. The glossy look of Lacobel combines seamlessly with its matt alter ego Matelac.

This product creates a striking color contrast that enhances the room with a stylish and sophisticated atmosphere.



Clearsight™ Ⅱ

Anti-reflective Glass Ideal for Outdoor Use

Clearsight™ is anti-reflective glass with special coating applied on both surfaces, thereby reducing light reflectance of the glass surface to 0.9%, whereas normallyis about 8.0%.

With improved coating durability, Clearsight™ Ⅱ can be used outdoor. Also, it can be processed into laminated glass.



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Production of Flat Glass -Float Process-

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Approx. Over 1,600℃

Approx. 1,100～1,300℃

1 Raw materials charge



2 Melting furnace



3 Fining furnace

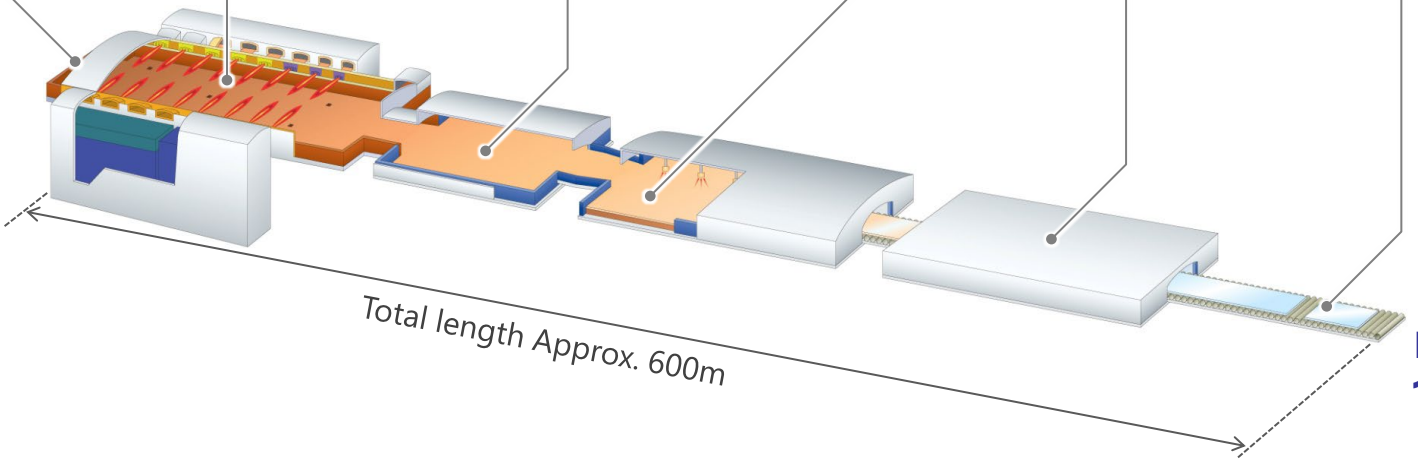
4 Float bath

5 Annealing



6 Cutting





Raw materials

- Silica sand
- Soda ash
- Salt cake
- Dolomite
- Limestone
- Feldspar

Glass size ;
Maximum 10m×3m

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Automotive

Automotive Glass

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	General name		Description of the product
Basic glass	Laminated glass		Glass with high safety and crime-prevention functions whose fragments do not scatter when it is broken and unlikely to penetrate thanks to adhesion of two sheets of glass that sandwich a film
	Tempered glass		Glass with improved strength and high safety by heating and rapidly cooling glass. When it is broken, fragments are grained.
High-function glass	Comfort	99% UV cut glass	Glass with a function to cut ultraviolet rays by about 99% to reduce long-term damage to the skin such as burns
		IR cut glass	Solar control glass that greatly cuts the wavelength range that we feel is the hottest among solar rays and has radio wave transmissiveness by inserting a special film between two sheets of glass.
			Solar control glass that has a function to reflect mainly infrared rays by coating the inner surface of laminated glass with a special film.
		Privacy glass	Glass that secures privacy as well as has high solar control performance thanks to the addition of colored components.
		Sound insulation glass	Glass that contributes to greater silence during driving by improving the sound insulation performance of laminated glass.
		Laminated side window	Glass that improves theft-prevention performance and sound insulation performance by using laminated glass for side glass.
		Light Control Glass	A special film is placed between two sheets of glass to freely control light transmission. In transparent mode, the glass provides a sense of openness, while in dimming mode, it provides privacy and blocks sunlight.
		Low-E Glass	Special Low-E coating for automotive use blocks solar heat and provides a cool and comfortable cabin temperature in summer and a warm and comfortable cabin temperature in winter by preventing heat inside the car from escaping outside.
	Eyesight improvement	Water repellent door glass	Door glass that improves visibility in the rain with high water repellency and durability thanks to highly reactive fluorine and silicone coating.
		Snow-melting/ Ice-melting front glass	Glass that melts snow and ice through energization by printing conductive ink (heating element) on the front glass.
		Electro-thermal defogging glass	Glass that defogs through energization by printing conductive ink (heating element) on the rear glass.
	Information communication	Printed glass antenna	Automotive antenna with excellent design and durability by casting conductive ink with glass by printing.
		Embedded DTV glass antenna	Digital TV (DTV) glass with excellent design and durability with a seal-type antenna sealed in the front glass.
		Glass for head-up display	Front glass with a function to display the speedometer, etc. on glass.
	Design	Encapsulated window	Glass with resin parts cast around glass.

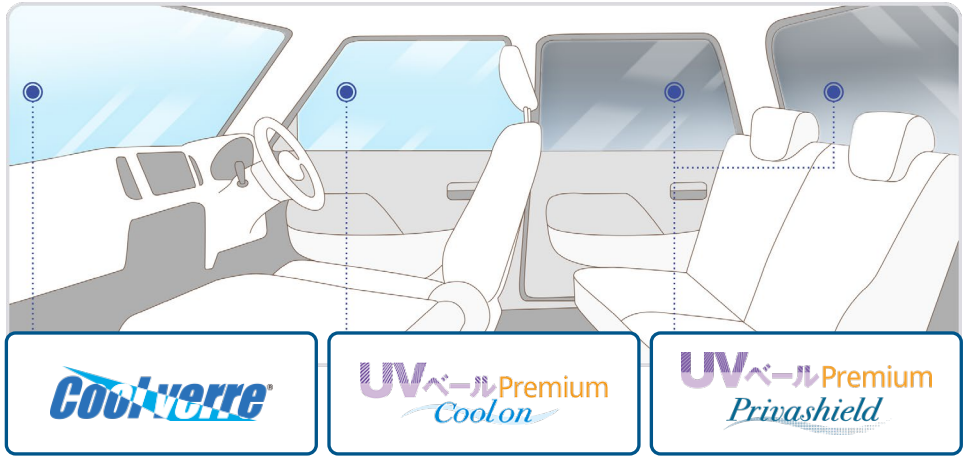
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UV Verre Premium™

Protects the skin of the driver, front seat passenger, and rear seat passengers from ultraviolet rays that fall from all directions in the car by approximately 99%*.

- In addition to the ultraviolet (UV) blocking function, which causes sunburn and blotches, it also has an infrared (IR) blocking function, which causes heat from the sun and a feeling of being frizzled, eliminating discomfort such as "it is hot inside the car" and "my arms feel frizzled while driving.
- By relieving the heat and enabling a lower air conditioner setting, the system also contributes to the reduction of CO_2 emissions by reducing the cooling load.



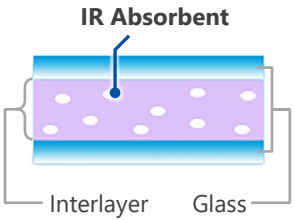
* AGC's survey data, based on ISO9050.

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Coolverre™

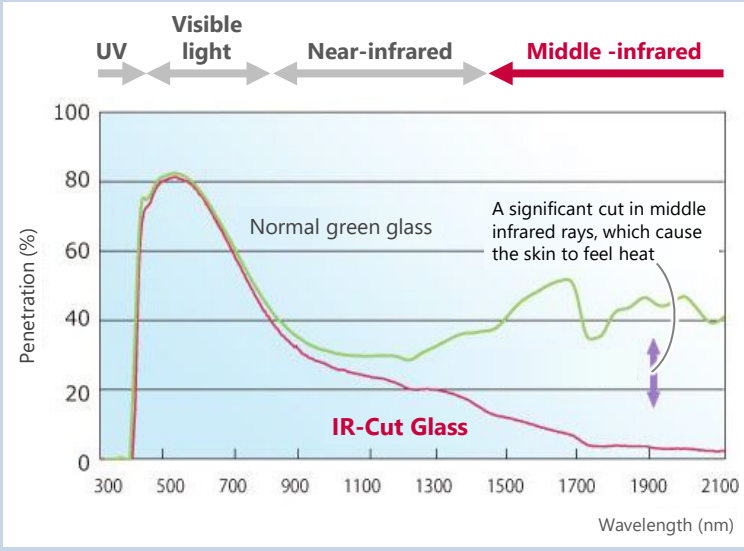
Increased comfort by cutting both IR & UV rays

UV absorbent kneaded into interlayer



- Efficiently blocks middle infrared rays (IR), which cause the skin to feel heat under the sun, and reduces irritations caused by direct sunlight.
- Provides greater interior comfort and improves fuel efficiency by controlling temperature increase inside the vehicle.
- Filters out approximately 99% of UV rays, which can cause aging of the skin
- Good radio-wave transmission

Comparison in the Penetration of Sunrays



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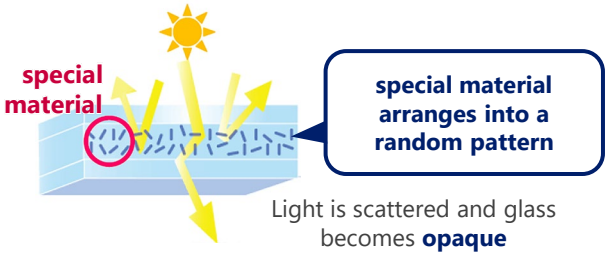
WONDERLITE™ Dx

Structure of laminated glass that sandwiches a special film. Realize a comfortable in-vehicle space according to the scene.

- World's fastest switch control
- Cuts ultraviolet rays by 99% in both dimming and transmission modes
- Cuts ultraviolet rays to protect the skin of occupants, and contributes to reducing CO_2 emissions by reducing cooling loads



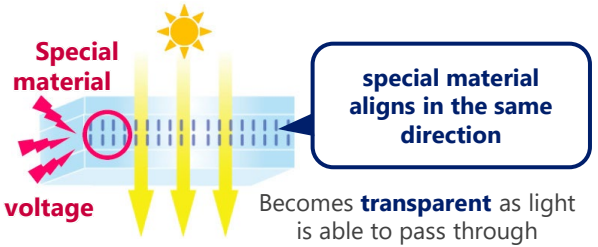
Mitigates the brightness of sunlight



Light is scattered and glass becomes **opaque**



Creates a sense of openness



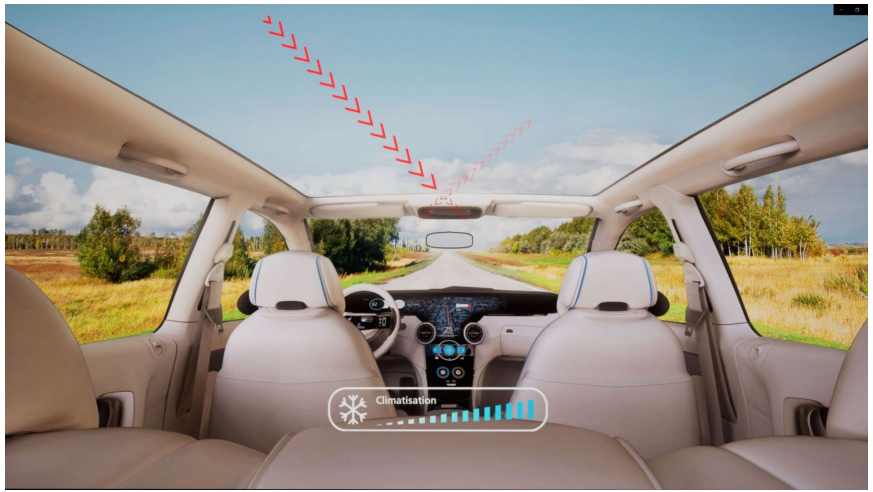
Becomes **transparent** as light is able to pass through

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Low-E Glass

Special Low-E coating for automotive use blocks solar heat and provides a cool and comfortable cabin temperature in summer and a warm and comfortable cabin temperature in winter by preventing heat inside the car from escaping outside.

- It reduces CO_2 emissions by reducing the air conditioning load and improving fuel economy, and contributes to extending the cruising range of EVs.
- When used on roof glass, it enables a shade-less setting, contributing to weight reduction of the vehicle body and securing head clearance (the distance from the top of the head to the ceiling when sitting in the seat).



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Specialty Glass for Chemical Strengthening

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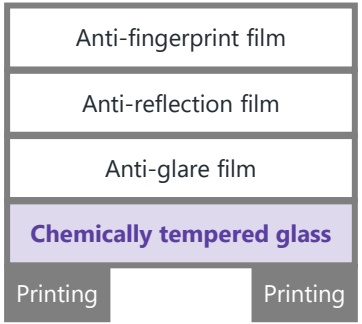
With exceptional strength and high-end appearance, chemically strengthened Dragontrail opens up new possibilities for automotive interior design.

- Strength and stability of Dragontrail contributes to safe and comfort driving experience.
- High scratch resistancy and various surface treatment options provide long lasting surface beauty, clear display image under all environment and smooth touch surface.

3D curved cover glass



Structure of ordinary cover glass



Value added interior with
safety + function + design

Strong & shiny





©AGC Inc.

Cover glass for car-mounted display

- **Track record of adoption:** Delivered more than 30 million sheets for over 100 vehicle models since the start of production in 2013

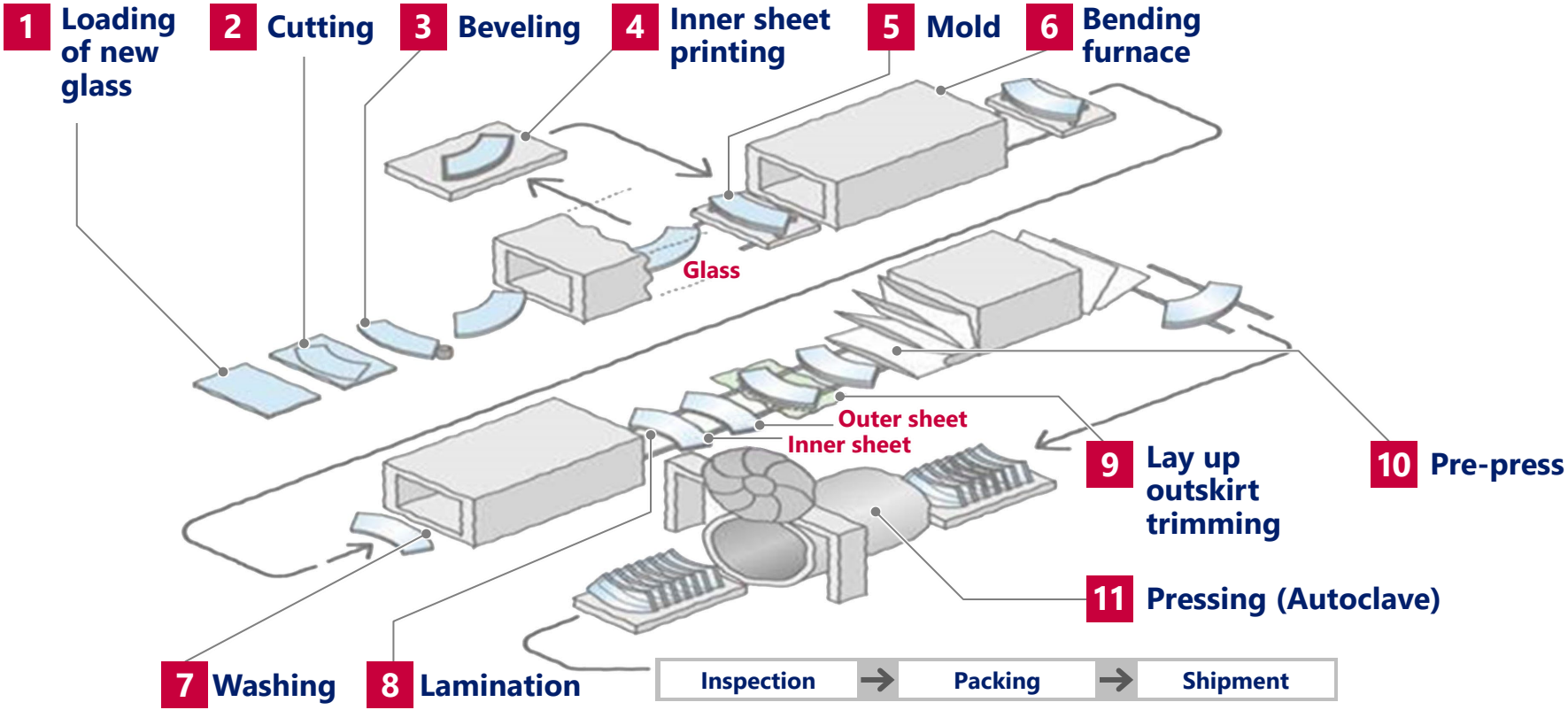


- **Major adoption cases** (excerpts)

Company	Audi	TOYOTA	General Motors
Vehicle model	Audi "**A8**"	LEXUS "**RX**"	Cadillac "**Escalade**"
Announced date	September 14, 2017	September 2, 2019	September 10, 2020

Production of Automotive Laminated Glass

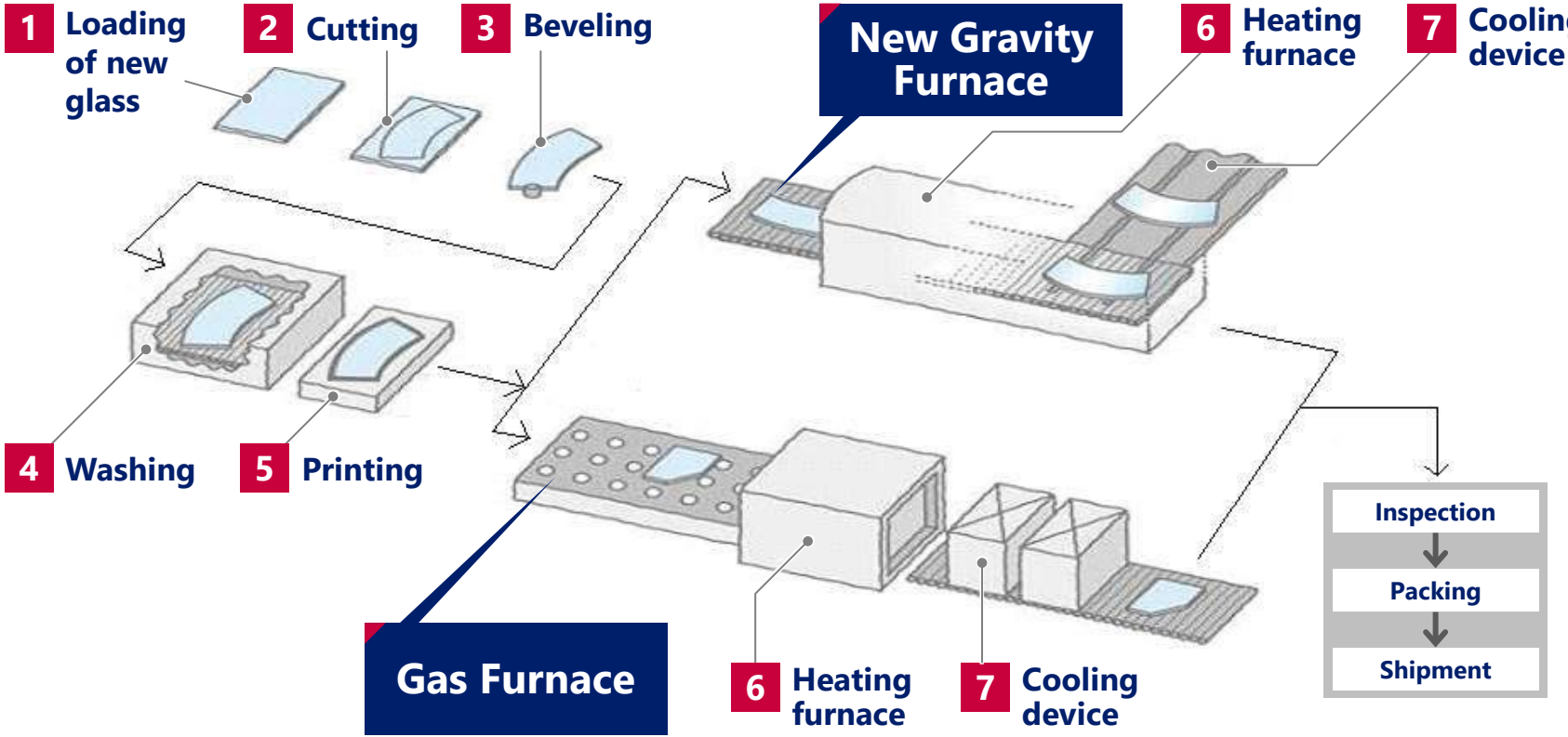
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Production of Automotive Tempered Glass

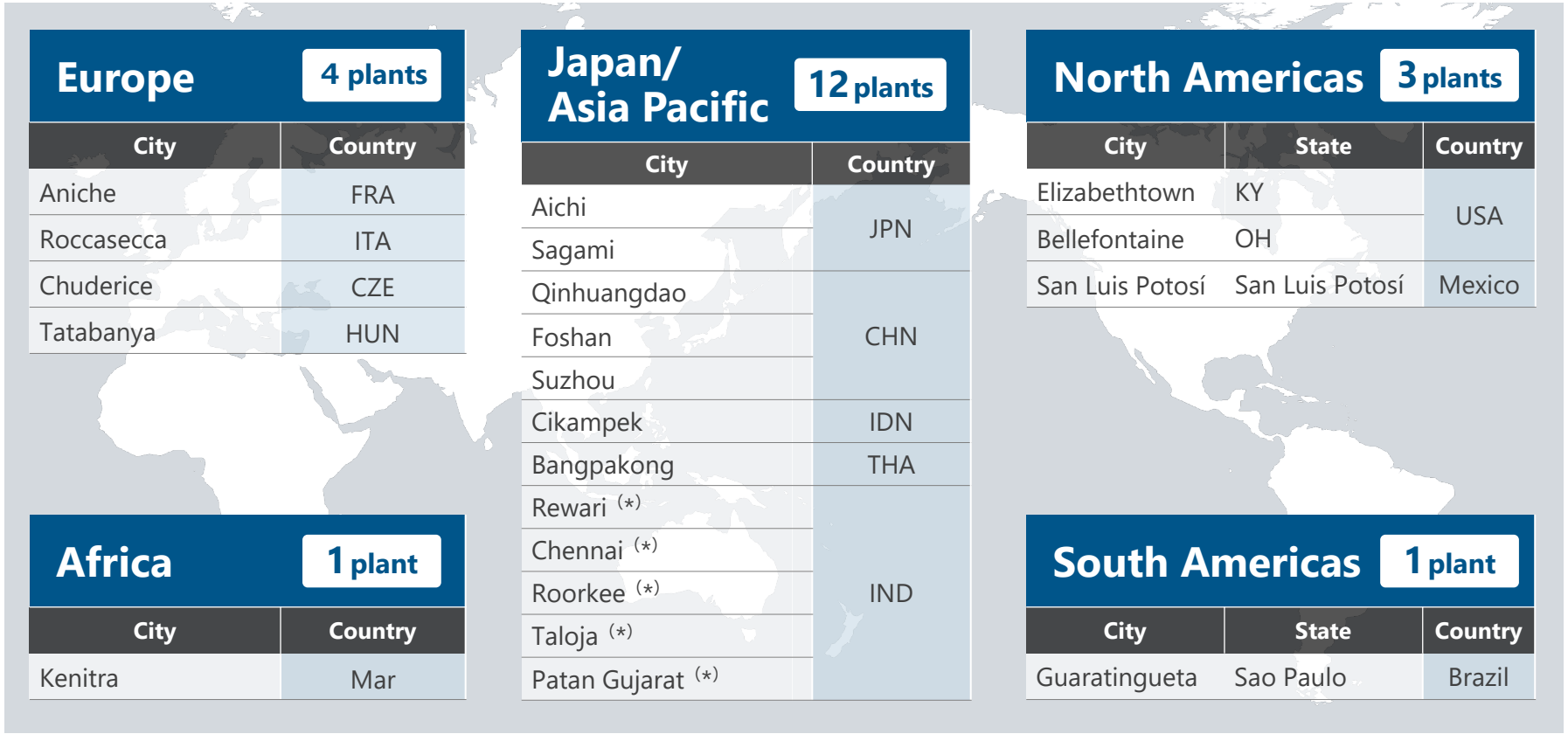
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Manufacturing Bases for Automotive Glass

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Europe — 4 plants

City	Country
Aniche	FRA
Roccasecca	ITA
Chuderice	CZE
Tatabanya	HUN

Africa — 1 plant

City	Country
Kenitra	Mar

Japan/ Asia Pacific — 12 plants

City	Country
Aichi	JPN
Sagami	
Qinhuangdao	CHN
Foshan	
Suzhou	
Cikampek	IDN
Bangpakong	THA
Rewari (*)	IND
Chennai (*)	
Roorkee (*)	
Taloja (*)	
Patan Gujarat (*)	

North Americas — 3 plants

City	State	Country
Elizabethtown	KY	USA
Bellefontaine	OH	
San Luis Potosí	San Luis Potosí	Mexico

South Americas — 1 plant

City	State	Country
Guaratingueta	Sao Paulo	Brazil

As of February 2023

(*) Equity method affiliate

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Electronics Segment : Display

Main Products

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General name	Product name	Product description
Glass Substrates for TFT-LCD／OLED Displays	**AN100**	Alkali-free aluminosilicate glass substrates manufactured by the float process. This display material has an exceptionally smooth and flat surface, excellent transparency and thermal resistance.
	AN Wizus™	Glass substrates with the world's lowest level of thermal shrinkage. Optimal for use in high-resolution display panels on smartphones and tablets, etc.
	AN Rezosta™	Large glass substrates for high-definition, large-screen displays with the industry's highest Young's modulus.
Specialty Glass for Chemical Strengthening	**Dragontrail™ series**	A specialty glass for chemical strengthening used as a cover glass for touchscreens on electronic devices such as smartphones, tablets, PCs, and automotive displays.
High-quality soda-lime glass	**AS2**	Glass for sensor substrates, various display devices, touch panels for smartphones and tablet PCs, and cover glass.

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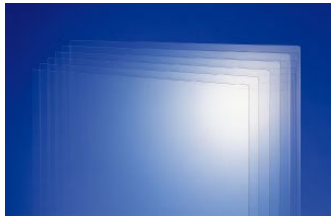
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AN100

AGC's Strengths

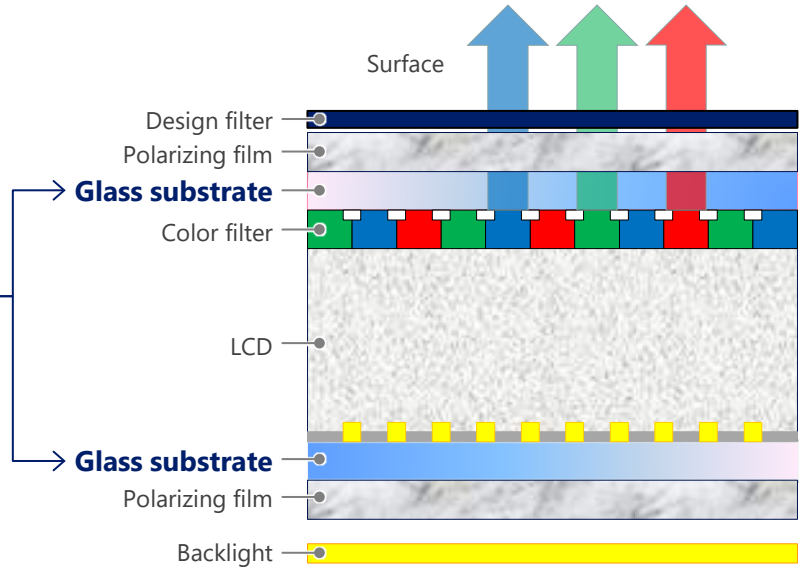
Stable production of aluminosilicate glass, which has transparency, smooth and flat surface, and excellent heat resistance.

TFT glass substrates: AN100



AN100 is alkali-free glass. It does not contain alkaline substances (soda ash) that are harmful to TFTs.

Cross-sectional view of LCD TV



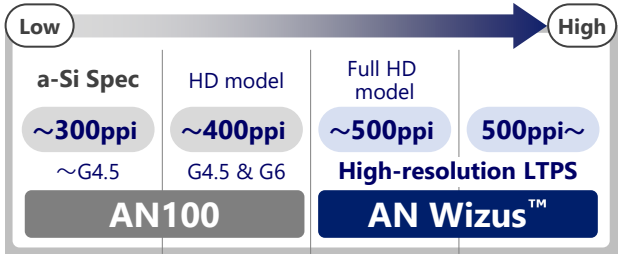
©AGC Inc.

AN Wizus™

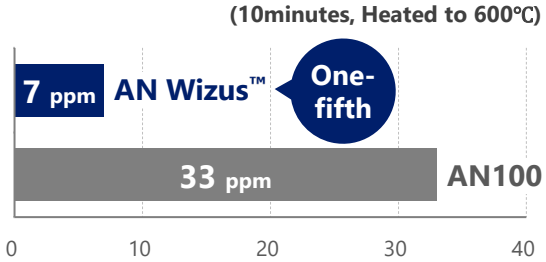
AGC's Strengths

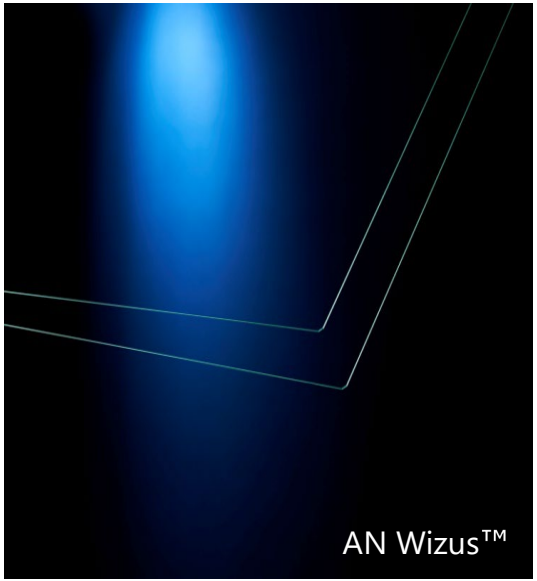
Optimal glass composition design for high-definition panels and production using the float method with a long slow cooling process suitable for producing low thermal shrinkage glass, resulting in overwhelmingly low thermal shrinkage.

Resolution



Thermal Compaction





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AN Rezosta™

AGC's Strengths

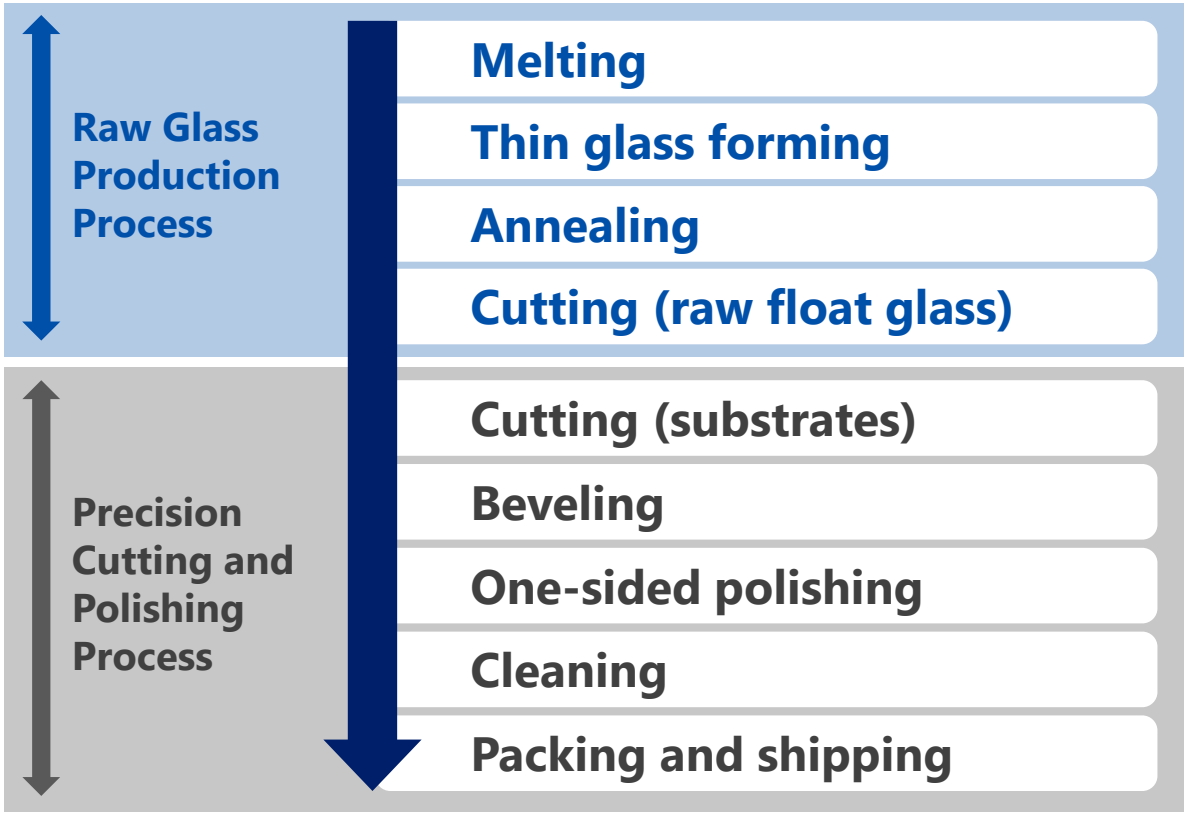
With the industry's highest Young's modulus, low thermal shrinkage, and capability of production of 8th-generation and larger area, "AN Rezosta™" will contribute to improving performance and productivity in the large, ultra-high-resolution, high-refresh-rate panels of its customers.



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Manufacturing Process for Glass Substrates for TFT-LCD／OLED Displays

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Float Process

(mass production, large size and low cost)

Inline Polishing Process

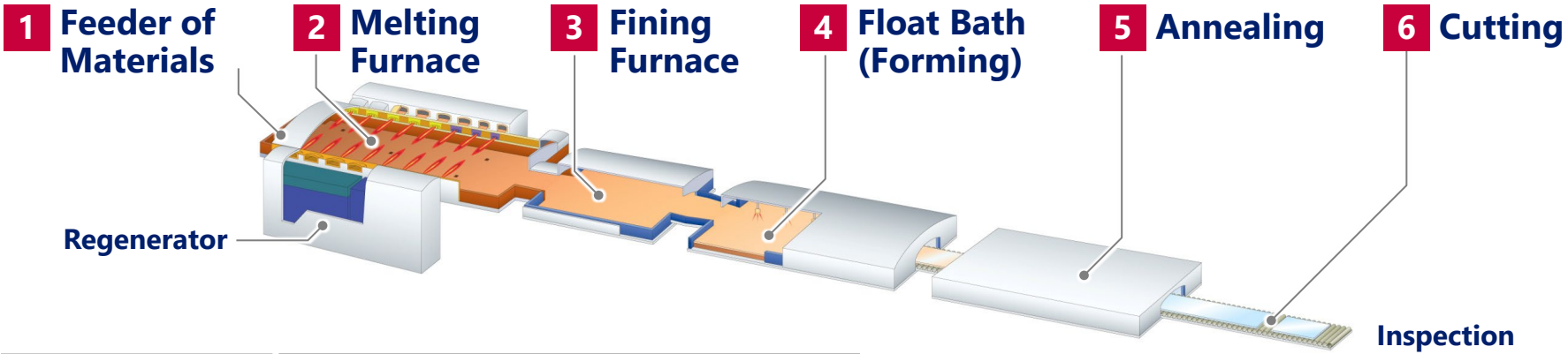
(very flat precision surface form)

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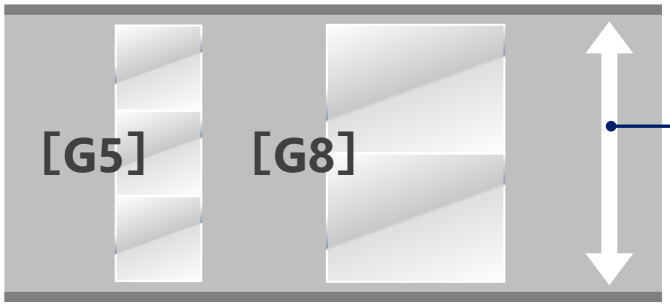
Production Method of Glass Substrates for TFT-LCD／OLED Displays

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Float Process



AGC's Float Process Cutting Image

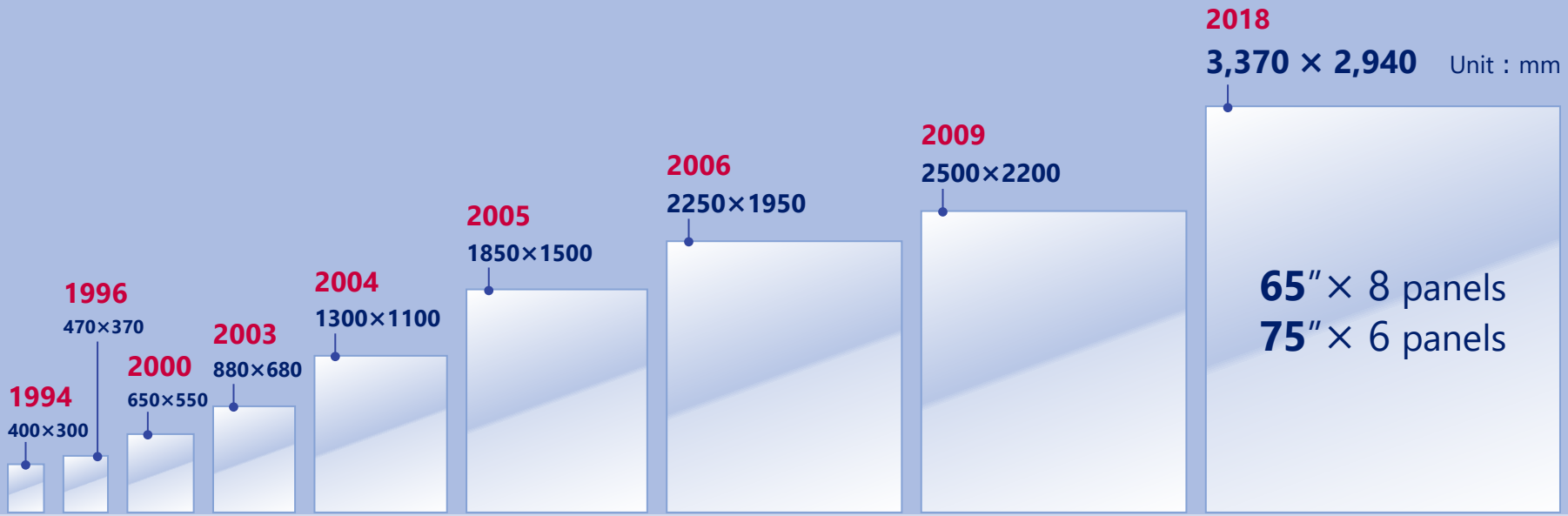


The float process has greater effective width, which allows the production of a multiple number of large-sized glass substrates at a time.

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Size Trend of Glass Substrates for TFT-LCD / OLED Displays

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Dragontrail™ series

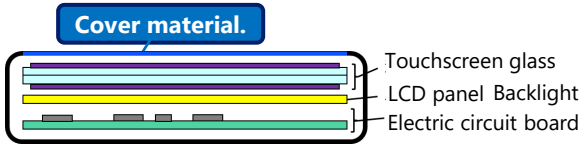
Dragontrail®

A specialty glass for chemical strengthening used as a cover glass for touchscreens of smartphones, tablet ,PCs and other electronic devices

AGC's Strengths

- Highly resistant to scratches and features a beautiful, pristine finish compared with resin.
- Dragontrail™ is manufactured by using the float method, a highly efficient production method as compared with fusion method and other manufacturing process.

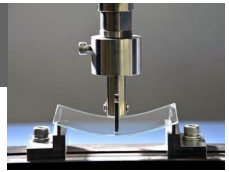
Structure of touchscreen devices



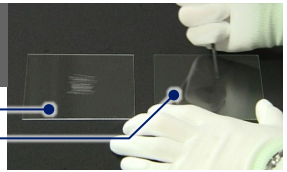
Characteristics of Dragontrail™ series

Strength test

three-point bend test on 1.1 mm-thick glass plate

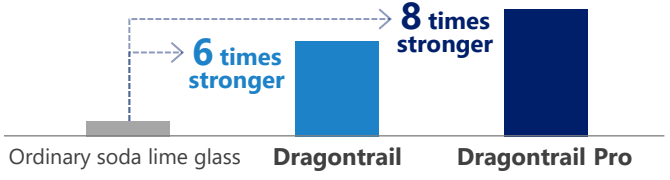


Abrasion Test



Strength Comparison of Dragontrail™ series





*Whencompared with glass of the same thickness

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High-quality Soda-lime glass

High-quality soda-lime glass "AS2" is used in various parts of display devices and electronic equipment.

It is manufactured by the float method, which is highly efficient in production, and is available in a variety of thicknesses from 0.23 to 1.1 mm.
We can meet the needs of a wide range of customers.



For TN/STN-LCD

As a substrate for LCD panels, AS2 contributes to the realization of high yields.



For cover glass

AS2 is used as tempering cover glass for general-purpose devices such as tablets, PCs, car-mounted display and industrial devices.



For touch sensor

Taking advantage of the thermal stability of glass, AS2 is used in touch sensors for car-mounted display.

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Manufacturing Bases

Product	Plants		Region
Glass Substrates for TFT-LCD／OLED Displays	■ Kansai Plant (Amagasaki factory)		Japan
	■ AGC Display Glass Taiwan		Taiwan
	■ AGC Fine Techno Korea	■ AGC Display Glass Ochang	Korea
	■ AGC Display Glass (Kunshan) ■ AGC Display Glass (Shenzhen)	■ AGC Display Glass (Keishu) ■ AGC Advanced Electronics Display Glass (Shenzhen)	China
Specialty glass for display applications	■ Kansai Plant (Takasago factory)		Japan
	■ AGC Flat Glass Protech (Shenzhen) Inc.		China
High-quality soda-lime glass	■ AGC Flat Glass Thailand		Thailand

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Electronics Segment : Electronic Materials

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Semiconductor-related materials （1）

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Synthetic Fused Silica Glass



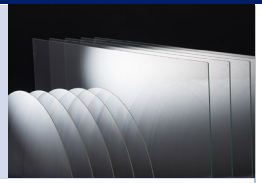
Ultra thin glass contributes to thinner and dervisified design for electronics devices with touch-panels

〈**Product Name**〉 Synthetic Fused Silica Glass AQ series

Applications
- Lens material for IC/LCD steppers/scanners
- Photo mask substrate
- Other optical material
- Glass wafer

Features
- High transmittance for deep ultraviolet rays with frequencies
- Excellent Compaction/Rarefaction and Polarization Induced Birefringence(PIB) characteristics
- Excellent heat resistance and durability
- Thermal expansion as low as 1/10 that of conventional glass
- High uniformity, Very high purity, very low metal impurity content
- Low OH group content
- Outstanding chemical resistance, Low dielectric loss

Glass substrate for semiconductor packaging



Glass substrate suitable as a support substrate for semiconductor package processes

〈**Product Name**〉 Glass substrate for semiconductor packaging

Applications
- Support glass for wafer-level packaging
- Support glass substrate for panel-level packaging

Features
- Compatible with wafer sizes such as φ300mm and panel sizes equivalent to 500mm square, etc.
- Supports high flatness and high smoothness
- Compatible with notches, markings, and non-contact/contact

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Semiconductor-related materials (2)

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Silicon Carbide



High purity, high strength, low thermal expansion Silicon carbide

〈**Product Name**〉 ROICERAM™-HS

Applications
- Material for semiconductor production equipment
- Material for LED manufacturing device
- Material for solar cell production equipment

Features
- The product has characteristics of high purity, high strength, low thermal expansion and excellent acid resistance and heat resistance.
- We have over 30 years of experience as a supplier of parts for semiconductor manufacturing furnaces mainly in high temperature process.

CMP Slurry



AGC offers slurry and polishing solutions for CMP processes

〈**Product Name**〉 CMP Slurry

Applications
- Front-end semiconductor process (wafer process)
- Various polishing for back-end process (package process)

Features
- Design and manufacturing of abrasive particles
- Accumulated technology of polishing materials for glasses
- Design and manufacturing of various analyses and chemical liquid
- Polishing material evaluation technology

EUVL mask blanks



Consumable components required for EUV lithography technology

〈**Product Name**〉 EUVL mask blanks

Applications
- EUV lithography

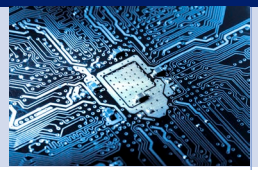
Features
- Producing high-purity substrates with technology and knowledge cultivated over many years
- We have developed a special polishing method to achieve ultra-high planarization through integrated production of substrate and polishing.
- Responds to customer requirements with optimal film design capabilities for high-definition applications and film formation technologies that reduce defects

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Semiconductor-related materials （3）

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CCL



CCL (Copper Clad Laminate)

〈**Product Name**〉 Series of Meteorwave

Applications

- High-speed communication
- Aerospace and defense systems
- Automotive millimeter-wave

Features

- Low dielectric loss, high reliability, and excellent thermal and mechanical properties
- Global production and sales support system
- Materials compliant with REACH and RoHS

Through Glass Vias



Glass substrate with through vias which enables 3D integration

〈**Product Name**〉 Through Glass Vias

Applications

- Circuit boards for semiconductor packages, glass interposers
- 3D glass IPD (Integrated Passive Devices)
- MEMS (Micro-Electro-Mechanical Systems) and sensor devices

Features

- **iniaturization**: The excellent flatness, shape stability, and fine pores of glass enable fine wiring.
- **High-speed performance**: Glass's high insulation and low dielectric loss tangent provide excellent high-frequency characteristics, enabling high-speed, large-capacity communication.
- **High reliability**: Alkali-free glass and a thermal expansion coefficient close to silicon make it ideal for semiconductor packages.
- **High productivity**: Compatible with sizes ranging from wafer to panel. ogy

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Optoelectronic materials （1）

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High refractive index glass



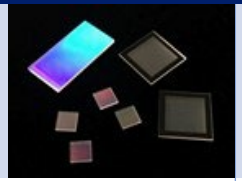
Glass substrates for next generation displays

〈**Product Name**〉 High refractive index glass

Applications
- Glass waveguide used on AR/MR smart glasses
- Glass waveguide for automotive head-up display

Features
- High refractive index
- Low absorption loss
- Flatness
- Smoothness
- Good thermal and chemical reliabilities

DOE・Diffuser



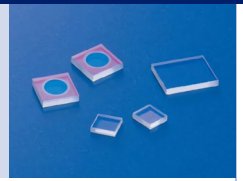
Micro-structured glass components for various consumer electronics and industrial laser applications

〈**Product Name**〉 Diffractive optical element(DOE)/Glass diffuser

Applications
- Sensing
- AR/VR/MR
- Projector, Optical communications, Lighting
- Laser beam shaping

Features
- One-stop service from design, prototyping through evaluation
- Accumulated skills and experience on DOE processing
- Small 0th order transmission, high efficiency, high durability
- Capabilities of very large FOV and high density dot projection
- Multi-level steps processing

Wave plate



An optical element capable of converting the vibration direction of light to any desired direction

〈**Product Name**〉 Wave plate

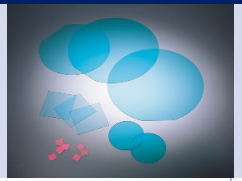
Applications
- Optical communication devices
- Optical imaging devices

Features
- Glass/Organic Thin Film/Glass Structure
- Patternable (Segmented Waveplate)
- Capable of Transferring to Arbitrary Polarization States
- Low-Order Waveplate
- Broadband Wavelength and Wavelength Selectivity
- Low Wavefront Aberration
- High Reliability (Compliant with GR-1221)
- Integration with Other Functions

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Optoelectronic materials （2）

IR cut filter



Visibility compensation filters for cameras

〈**Product Name**〉 NF Glass Filter

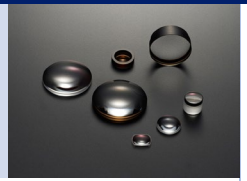
Applications

- Digital single-lens reflex camera, compact camera, smartphone camera, cameras for automobiles, surveillance camera, etc.

Features

- excellent image quality with absorption type
- It reduces ghosting and flares which have a bad influence on the image quality.
- Incident angle dependence of spectral transmittance is small
- integrated production system in a range from glass melting to forming/processing
- The variety of lineups are available so as to meet the spectral characteristics by the customers' requirements, such as anti-reflection coating and UVIR cut film

Aspherical Glass Lenses



Aspherical Glass molded lenses improving the optical performance of precise equipment

〈**Product Name**〉 Aspherical Glass Molded Lenses

Applications

- Digital single lens reflex cameras (DSLR), mirrorless interchangeable lens camera, compact camera, surveillance camera, cameras for automobile
- Projectors
- Sensing equipment

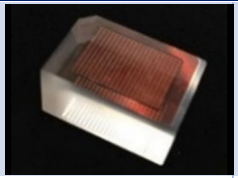
Features

- Lens shape: biconvex, meniscus (convex and concave), biconcave, toroidal, etc.
- Lens type: single lens and cemented lens (Doublet, Triplet)
- Size: φ1mm to φ48.5mm or so
- Special processing: I cut, D cut, and rectangular cut, etc.

Optoelectronic materials （3）

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Micro Lens Array



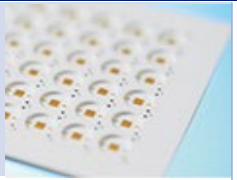
Optical Elements with one or two dimensionally arrayed small lenses on a glass substrate

〈**Product Name**〉 Micro Lens Array

Applications
- CCD/CMOS image sensors
- Optical communication devices, optical interfaces, CPO
- Digital cameras
- High-brightness LEDs, organic LEDs

Features
- Submicron-level high-precision lens formation
- High-accuracy lens arrangement with various spacings
- Capable of manufacturing both single-sided and double-sided arrays
- Anti-reflection coatings and other layers can be applied via surface coating
- Formation of fine lens arrays with lens pitches of several tens of micrometers
- Customizable functional enhancements designed to meet specific applications.

Glass Ceramics Substrate



Products for contributing brightness and high-output of LED and semiconductor laser

〈**Product Name**〉 GCHP™

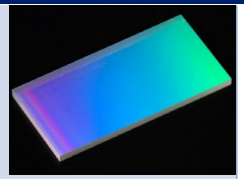
Applications
- Visible LED, IR LED, UV LED
- Semiconductor Laser
- headlamp

Features
- Integrated development from glass material to finished products
- High optical reflectance over a wide range of spectrum (UVC-NIR)
- High heat release efficiency
- Package size reduction capability by 3D circuit integration (multilayer)
- High reliability realized by glass (no color change)
- High hermetic properties of dense glass
- Prevention of solder cracking by glass

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Optoelectronic materials （4）

Transmission grating



An element that diffracts waves such as light

〈**Product Name**〉 Transmission Diffraction Grating

Applications

- Optical communication
- LiDAR
- Attenuator
- Optical spectrum analyzer

Features

- Propose high-efficiency designs based on customer requirements
- Design of ultra-fine pitch gratings at the submicron level
- Capable of designing low PDL (Polarization Dependent Loss) and low WDL (Wavelength Dependent Loss)
- Multi-stage processing
- Compliant with Telcordia (GR-1221) standards
- Customizable sizes based on requirements (from several mm^2 to several tens of mm^2)
- Low thermal expansion coefficient due to the use of synthetic quartz

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Optical Thin Film

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Antireflection Film



Reduce the surface reflection and improves the permeation rate

〈**Product Name**〉 Antireflection Film

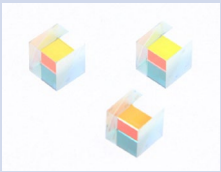
Applications

- Products using lenses, processing equipment, medical measuring equipment, etc.

Features

- Single- or multi-layer dielectric films are formed on the surface of optical materials to prevent light reflection on the surface and improve transmittance.
- Suppresses ghosting in optical elements.

Beam Splitter



Select an arbitrary wavelength and able to separate polarization components

〈**Product Name**〉 Beam Splitter

Applications

- Semiconductor and LCD lithography equipment, laser interferometers, measurement equipment, etc.

Features

- Reflects a portion of the single-incident light and also transmits a portion of the single-incident light. Also capable of separating p- and s-polarization components.
- Two types: cube and plate

Interference Filter (Dichroic Mirror)



Adjustment of permeation and reflection depending on the incidence angle is possible.

〈**Product Name**〉 Dichroic Mirror/Filter

Applications

- Color TV cameras, LCD projectors, color photo enlargers, color fax machines, lighting, other optical systems, etc.

Features

- AA multilayer film of dielectrics, each with a different refractive index, separates light in two or more wavelength regions. This action makes it possible to extract specific colors.

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Consumer business

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Tissue Culture Products



Reliable "IWAKI" Brand Plastic Products for Cell Culture

〈**Product Name**〉 Tissue Culture Products

Applications
- Drug Discovery Research
- Regenerative Medicine Research
- Various Studies in Cell Biology

Features
- Strict Quality Control
- In Addition to Standard Products, Unique Products with Specific Functions Are Available
- Culture Vessels with Various Culture Surfaces
- Culture Vessels Such as Dishes and Plates Are Designed for Easy Handling

Laboratory Glass Ware



High-quality heat resistant glass ware for Laboratories

〈**Product Name**〉 Laboratory Glass Ware

Applications
- Chemical Analysis
- Quality Control
- Environmental Measurement

Features
- Low-expansion borosilicate glass material with excellent heat resistance, chemical resistance, and transparency
- A lineup of glass products in a wide variety of shapes based on the accumulation of dissolution and processing technologies over many years

Heat Resistant Glassware



Heat resistant glassware usable everyday for food preparation, cooking, dining and storage

〈**Product Name**〉 iwaki Heat Resistant Glassware

Applications
- Kitchenware
- Tableware

Features
- Excellent heat resistance allows cooking in ovens and microwaves, and use in dishwashers and dryers.
- Excellent weather resistance, making it easy to clean and resistant to odor transfer.
- Excellent chemical resistance, unaffected by acids, alkalis, and other food components.

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Other

Glass Frit,Glass Pastes and Glass Lid for Hermetic Sealing



High-performance materials for sealing, coating, and bonding in the electronics field

〈**Product Name**〉 Glass Frit Paste/Preform & Low-temperature Hermetic Sealing Materials

Applications
- Electronic Devices (Displays, Printers, Sensors, Multilayer Circuit Boards, Packages)
- Electrochemical Devices (Fuel Cells, Solar Cells, Secondary Batteries, Various Types of Batteries)
- Semiconductor-related Applications

Features
- Available in Various Forms: Powder, Paste, and Molded Shapes
- Wide Range of Glass Options: Offering glass suitable for a broad range of firing conditions and thermal expansion coefficients, selected from over 400 types, including bismuth, borosilicate, vanadium, and phosphate glasses
- Precise Composition Control and Customized Development Support

Glass substrate for anodic bonding



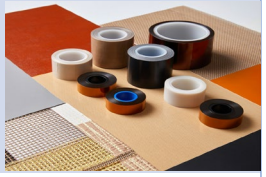
Glass with an expansion coefficient that is very close to silicon

〈**Product Name**〉 SW glass substrate

Applications
- Wafer-level Packages for MEMS Substrates
- Sensors: Pressure Sensors, Acceleration Sensors, Various Automotive Sensors, Various Industrial Sensors
- LD/UVC-LED Packages
- Various Optical Applications

Features
- Extremely Close Thermal Expansion Coefficient to Silicon Across a Wide Temperature Range
- Three Product Lineups Available for Selection Depending on the Application
- Metal Coating on Substrates, Through-hole Etching, and Counterbore Processing Also Available

Industrial PTFE Composite Materials



PTFE Fabrics, Belts, Silicone Tapes, Polyimide Composites

〈**Product Name**〉 Industrial PTFE Composites

Applications
- Semiconductors
- Architectural Membranes
- Food Processing

Features
- Excellent Heat Resistance, Non-stick Properties, and Chemical Resistance

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Major manufacturing Bases

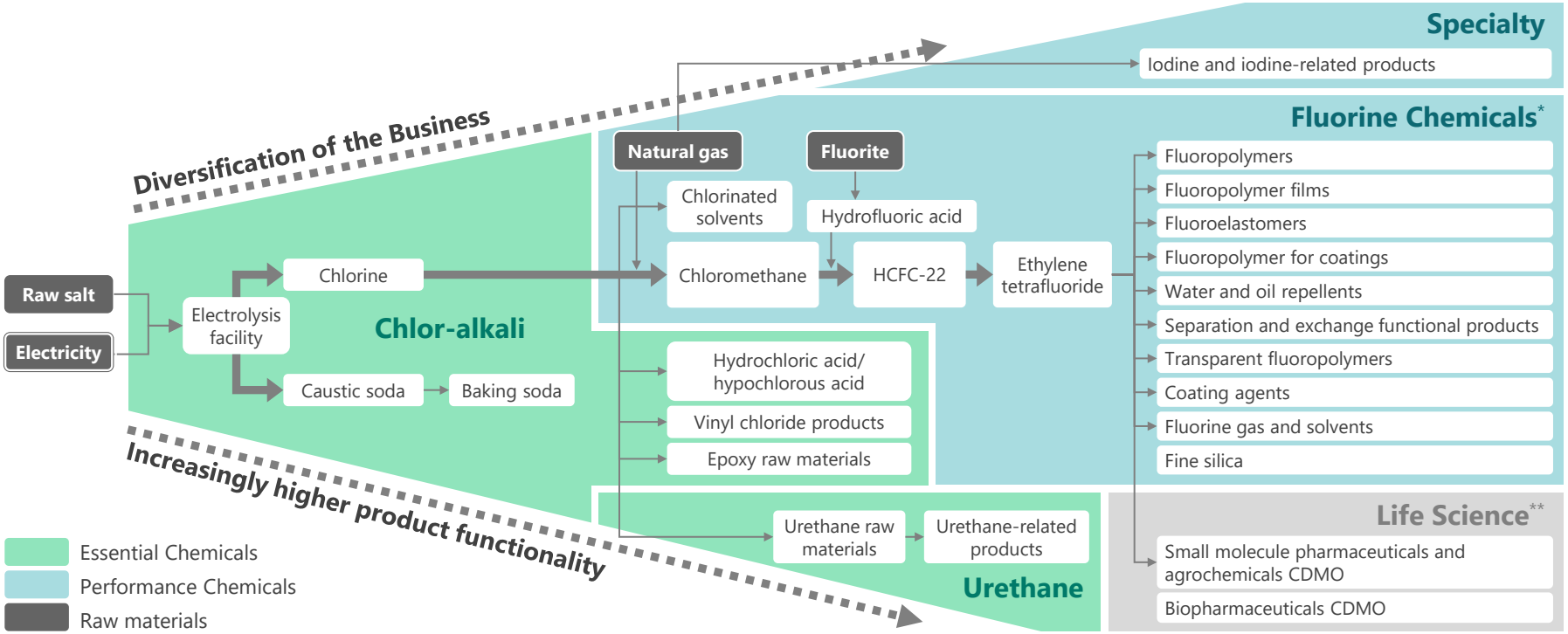
Product	Plants	Region
Semiconductor process materials	■ Kansai Plant (Takasago factory) ■ AGC Seimi Chemical ■ AGC Electronics	Japan
	■ AGC Electronics America	USA
Optical materials for cameras	■ AGC Techno Glass ■ AGC Micro Glass	Japan
	■ AGC Micro Glass (Thailand)	Thailand
Glass-ceramics substrate for high-power LED/LD lighting	■ AGC Electronics Taiwan	Taiwan
Optical Coating Products	■ Optical Coatings Japan	Japan
Glass frit/paste	■ AGC Electronics	
Tissue Culture Ware	■ AGC Techno Glass	
CCL（Copper Clad Laminate）	■ AGC Multi Material America	USA
	■ AGC Multi Material Singapore	Singapore
CCL（Copper Clad Laminate） Industrial PTFE composite Derivatives	■ AMMK	Korea

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Chemicals : Essential Chemicals Products

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Chemical Chain of AGC's Chemical Business



* Includes some products not made from fluorine ** Became a stand-alone segment after 2022

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Essential Chemicals Products

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General name	Product usage
Caustic Soda	Chemicals, chemical fibers, paper/pulp, etc.
Caustic potash	Chemicals, potassium carbonate raw materials
Potassium carbonate	Food additives, detergent feedstock
Hydrochloric acid	Inorganic chemicals, steel, and chemical seasoning
Sodium hypochlorite	Bleaching pulp and textile, and sterilizing tap water
Liquid chlorine	Bleaching pulp and sterilizing tap water
Vinyl chloride monomer	Polyvinyl chloride raw materials (VCM)
Polyvinyl chloride	Water supply pipes, construction materials (window sashes, wallpaper, flooring materials, etc.), electrical cable covers, furniture, food packaging, greenhouses
Trichloroethylene, Perchloroethylene	Industrial cleaner, dry-cleaning solvent, fluorinated gas feedstock
Methyl and ethylene chloride	Silicone resin, industrial cleaner, extracting solvent for pharmaceuticals and agrochemicals, paint stripper
Chloroform	Raw materials for fluorochemical products
Epi-chlorohydrin	Epoxy resin
Propylene oxide	Raw materials for propylene glycol, dipropylene glycol and polyols, raw materials for non-ion surfactants
Propylene glycol	Raw materials for unsaturated polyester resins and plasticizers, surfactants, antifreeze and coolants, food additives, perfume and tobacco, toothpaste, cosmetics and medical supplies
Polyols	Rigid and flexible polyurethane foams, coating, adhesives and sealant, elastomers
Sodium bicarbonate	Food additives, pharmaceuticals, bath salts, industrial applications and animal feeds

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Caustic Soda, Polyvinyl chloride

- Caustic soda and chlorine are produced from electrolysis of salt.
- Caustic soda is used in a wide range of application as a typical alkaline product for industrial use.
- Chlorine is processed into a variety of products, such as polyvinyl chloride(PVC).



Crude salt, raw material for chlor-alkali products.



Electrolysis system, using AGC's ion-exchange membrane (Flemion™)

Caustic soda



- **Paper/pulp**
- **Alumina refining**
- **Soap / detergent**
- **Sewer neutralization**

Polyvinyl chloride



- **PVC Pipes**
- **Electric cable jackets**
- **Window sashes**

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Polyurethane products

- Polypropylene glycol (PPG) is an indispensable main raw material for producing polyurethanes.
- We have an integrated production system from propylene oxide(PO), the raw material for PPG, to polypropylene glycol (PPG).
- AGC also produces modified silicones with high functionality.
- AGC also offers environment-friendly products, such as PPG suitable for water foaming which enables molding without usage of CFCs, and PPG which can be used with HFO, a new environmentally friendly refrigerant.

PO

- Raw material for PPG
- surface-activating agent



PG

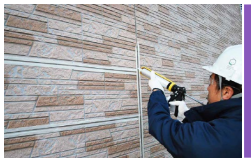
- Medical
- Cosmetics
- Food additives

PPG EXCENOL, PREMINOL

- Foaming application
  - ・rigid foam (insulator material)
  - ・flexible foam(seat cushion)
- Non-foaming application
  - ・Coating Adhesive Sealant Elastomer



high function PPG



Modified silicones
EXESTAR

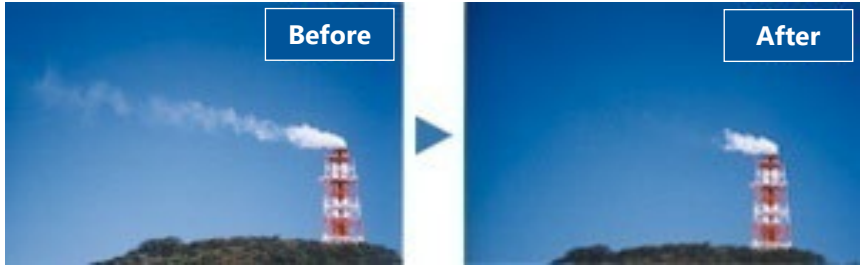
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Sodium Bicarbonate

- Sodium bicarbonate is used in various fields, ranging from home products, medical products to advanced technologies.
- As the largest sodium bicarbonate manufacture in Japan, AGC offers a fine selection of grades. AGC is also committed to develop new products to meet wide range of needs.

Applications

Industrial cleaning	Blast cleaning media, Chemical cleaning new system
Flue gas treatment	Flue gas neutralization
Pharmaceuticals	Stomach medicine (antacid), dialysis
Home products	Cleaners, bath salts, deodorant
Food products	Baking powder, PH control chemicals
Agricultural / Livestock industries Civil engineering / construction	Formula feed ingredients, Animal pharmaceuticals, Agrochemicals ingredients Liquid glass soil hardening agents
Industry	Fire extinguisher, metal surface treatment agents, neutralizer



Sodium bicarbonate can also be used as a highly reactive neutralizer for exhaust gas treatment, efficiently removing acidic components in exhaust gas such as HCL, SOX, sulfuric acid mist, and etc..

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Chemicals Manufacturing Bases of Essential Chemicals

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Chlor-alkali & urethane

General name	Production base	Country
Caustic Soda	■ Chiba plant ■ Hokkaido Soda ■ Kashima plant	Japan
	■ P.T.Asahimas Chemical	Indonesia
	■ AGC Vinythai	Thailand
Caustic potash	■ Chiba plant	Japan
	■ AGC Vinythai	Thailand
Potassium carbonate	■ Chiba plant	Japan
	■ AGC Vinythai	Thailand
Hydrochloric acid	■ Chiba plant ■ Hokkaido Soda ■ Kashima plant	Japan
	■ P.T.Asahimas Chemical	Indonesia
	■ AGC Vinythai	Thailand
Sodium hypochlorite	■ Chiba plant ■ Kashima plant	Japan
	■ Hokkaido Soda	Japan
	■ P.T.Asahimas Chemical	Indonesia
	■ AGC Vinythai	Thailand
Liquid chlorine	■ Chiba plant ■ Kashima plant	Japan
	■ P.T.Asahimas Chemical	Indonesia
	■ AGC Vinythai	Thailand

General name	Production base	Country
Vinyl chloride monomer	■ Keiyo Monomer	Japan
	■ P.T.Asahimas Chemical	Indonesia
	■ AGC Vinythai	Thailand
Polyvinyl chloride	■ P.T.Asahimas Chemical	Indonesia
	■ AGC Vinythai	Thailand
	■ AGC Chemicals Vietnam	Vietnam
Trichloroethylene, Perchloroethylen	■ Chiba plant	Japan
Methyl chloride	■ Kashima plant	Japan
Methylene chloride	■ Chiba plant	Japan
Chloroform		
Epi-chlorohydrin	■ Kashima Chemical Co., Ltd.	Japan
	■ AGC Vinythai	Thailand
Propylene oxide	■ Kashima plant	Japan
Propylene glycol		
Polyols		
Sodium bicarbonate		

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Chemicals : Performance Chemicals Products

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Products of Performance Chemicals

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General name "Product name"		Product usage
Fluoropolymers "Fluon® ", "Fluon+™"		Tubing, sealing, wire insulation, films, filaments, lining, etc. for various industries
Fluoroelastomers "AFLAS®"		Wire insulation, high performance sealing materials for semiconductors, food manufacturing, oil & gas application, etc.
Fluoropolymer film "Fluon® ETFE FILM" Green house film "F-CLEAN®"		Mold releasing films, roofing and architectural facades, interior finishing and photovoltaics Green houses
Transparent Fluoropolymers "CYTOP"		Optical materials, semiconductor processing, hi-performance
Fluoropolymer for coatings "Lumiflon®"		Architectural structures, bridges and aircrafts
Ion-exchange membranes	FORBLUE™ FLEMION	Ion-exchange membrane for salt electrolysis (for caustic soda/caustic potash/chlorine production)
	FORBLUE™ S-SERIES	Water electrolysis, redox flow battery, Ion-exchange membrane for wide variety of electrolysis and electrodialysis processes
	FORBLUE™ SELEMION	Acid recovery, wastewater reclamation, desalination and concentration, desalination of groundwater
	FORBLUE™ sunsep	Dehumidification and humidification of various gases
Fluorine-based electrolyte polymer for fuel cells "FORBLUE™ i-series"		Material for power-generating system of fuel cell vehicles
Fluorinated gases and solvents "AMOLEA™1234yf", "AMOLEA™1224yd"		Refrigerants for freezer/refrigeration equipment, and air conditioners, working fluid, foaming agent
Fluorinated solvent "AE-3000", "AC-6000" "AMOLEA™AS-300"		Cleaning, rinsing and drying agents, dewatering, solvents, dispersant, heat transfer fluid
Fluorocoating agents "SFCOAT™"		Damp-proof coating for electronics devices, oil barrier, anti-flux migration, anti-resin adhesion
Fine silica RESIFA™ M.S.GEL™, RESIFA™ SUNSPHERE™, RESIFA™ SUNLOVELY™		Liquid chromatography packing agent, cosmetic materials

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Performance Chemicals : Fluoropolymer ETFE

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- Fluoropolymers **with ease of forming and fabrication while keeping the excellent characteristics of** fluorine. **Widely used in diverse and specific industrial fields**, namely transport equipment, electronics, construction, and energy.
- Extrusion molding, injection molding, and powder coating are possible. **Used in severe usage environment where thermal resistance, chemical durability, insulation, etc. are required** such as wire coatings, tubes, and coating materials.
- With ETFE, film fabrication is also easy. **Used as a mold-releasing film for membrane structure materials and other various fields**

Materials



Pellets



Powders

Fabrication materials



Wire coatings



Lining/coating products



Transfusion tubes



Transfusion tubes



Membrane structure material/ Mold releasing film

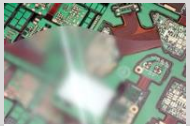
Industrial fields



Automobile



Train



Semiconductors



Aircraft



Electronic equipment



Energy



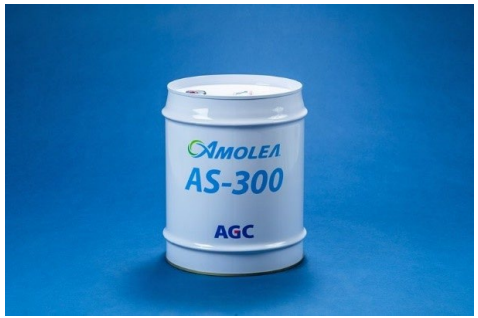
Membrane structures



Protected horticulture

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Environmentally friendly refrigerant & solvent AMOLEA™ series





- AGC's AMOLEA™ is a solvent and refrigerant brand developed under the concept of dramatically reducing Global Warming Potential (GWP) while maintaining conventional performance.
- We have the following three products for each application.

General name	Name of the Product	Applications
New environmentally-friendly refrigerant	**AMOLEA™ 1234yf**	refrigerant for mobile air-conditioners, refrigerators
New environmentally-friendly refrigerant	**AMOLEA™ 1224yd**	refrigerant for turbo chillers, working fluid, foaming agent
Low GWP Fluorinated Solvent	**AMOLEA™ AS-300**	Cleaning, solvent

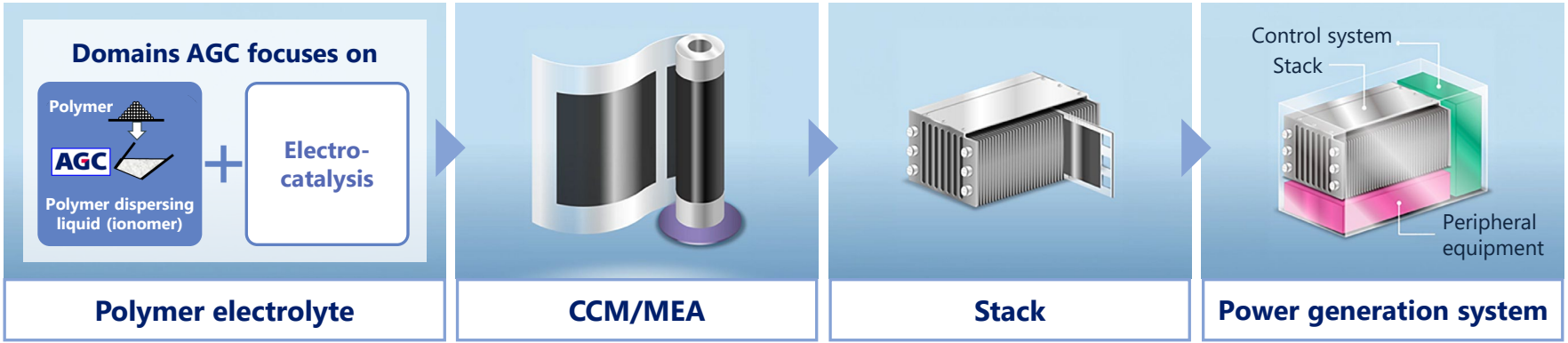
- (American Society of Heating, Under-development AMOLEA 1123 mixed refrigerant is a most promising candidate as a next-generation refrigerant which offers high freezing/heating capacity and low GWP at the same time. It is AGC's unique refrigerant, which is promising to lead market growth mainly for air conditioners for residential, commercial and electric vehicles.

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Fluorinated electrolytic polymer for fuel cells (PEMFC ionomer)

- Supplying **fluorine-based electrolyte polymer (PEMFC ionomer) for fuel cell membranes**, an essential component of fuel cells.
- Demand increase accelerated due to the spread of fuel cell vehicles (FCV) and technological development aimed at realizing a hydrogen society.
- AGC achieved No.1 position based on the excellent performance for high power generation and durability.

	Polymer electrolyte	CCM/MEA	Stack	Power generation system
For membranes	AGC Material manufacturers	■ MEA manufacturers ■ Automotive manufacturers	■ Automotive manufacturers etc.	■ Automotive manufacturers etc.
For electrodes	AGC Material manufacturers			



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Fluon+™ EA-2000

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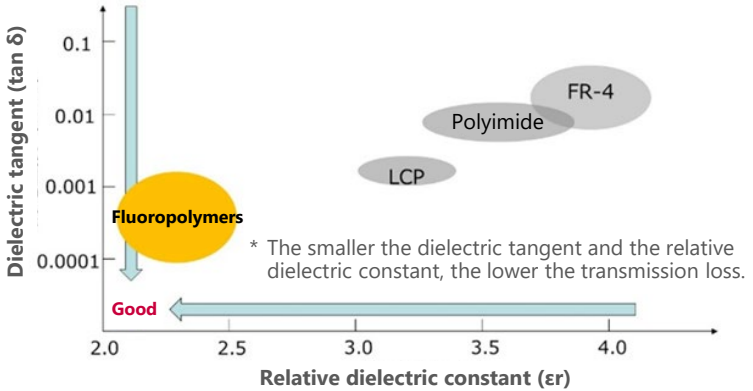
■ Adhesive fluoropolymer for next generation high speed communication.

- Fluon+™ EA-2000 is a low transmission loss fluoropolymer featuring low dielectric constant and low dielectric loss tangent
- Good adhesion and dispersibility that overturns the concept of existing fluoropolymers.
- Printed circuit boards using this product **can reduce transmission loss by 30% or more compared to existing materials.**

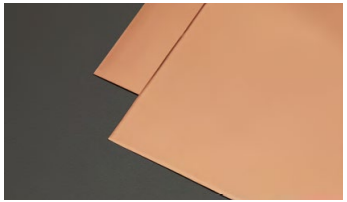
Printed circuit board/flexible printed circuit board applications suitable for Fluon+™ EA-2000

Automotive radar
Flexible printed circuit boards for 5G smartphones
millimeter-wave devices

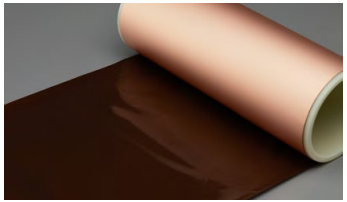
Dielectric constant and dielectric tangent



Source: Edited by AGC based on data from Nikkei Electronics August 2017 issue



Fluon+™ EA-2000/ PPE laminated CCL



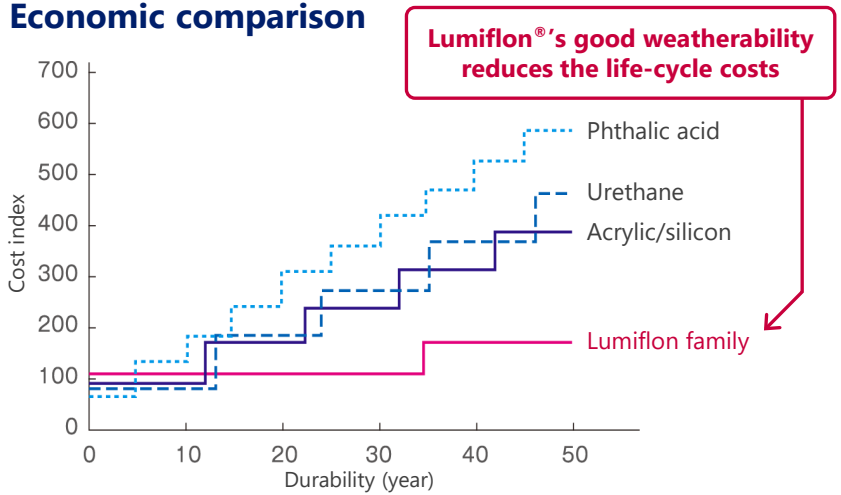
Fluon+™ EA-2000 coat RCC

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Fluoropolymer for Coatings Lumiflon®

- Lumiflon® is the world's first solvent-soluble fluoropolymer for coatings, developed by AGC's fluorine technologies, commercialized in 1982.
- Chemically-stable fluoropolymer demonstrates greater weatherability than conventional coating materials.
- Proven credibility based on more than 40 years of experience, used for over 200,000 buildings and facilities.

Economic comparison



Excellent weatherability



Bridge pier without Lumiflon®

(16 years-old)



Bridge pier with Lumiflon coating

(21 years-old)

Lumiflon® is used for:



Ping An Finance Centre,

Shenzhen



Akashi Kaikyo Bridge

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Silica

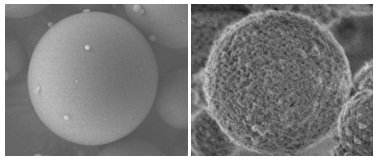
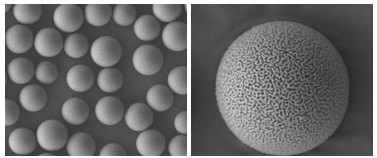
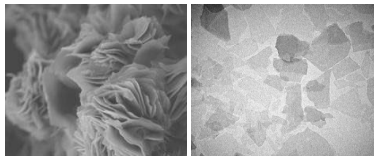
- AGC Si-Tech, as a professional in the silica business for long years, offers silica products suitable for various applications including cosmetics, industry, and pharmaceutical production.
- Silica is attracting attention as a material that contributes to the reduction of environmental risks since it is a material that originally exists in nature. For example, contributes to various social problem-solving, such as being expected to serve as an alternative to microplastics with concerns about their impact on the ecosystem due to their discharge into the ocean, and being utilized for semiconductor inorganic fillers.

RESIFA™

Concept of the new brand RESIFA™

Recycle	(to circulate nature)
+	
Silica	(with silica)
+	
Facilitate	(contributing to the realization of Customers' dreams)

RESIFA™ Product Lineup

SUNSPHERE™	M.S.GEL™	SUNLOVELY™
		
usage	**usage**	**usage**
■ Fillers for cosmetics and antiperspirants ■ Fillers for resins and films ■ Carriers for functional materials ■ Catalyst carriers	■ HPLC column packing material ■ Purification of pharmaceuticals and natural products ■ Carriers for functional materials	■ Binders for coatings ■ Hydrophilic fillers ■ Functional particulate binders

Manufacturing Bases of Performance Chemicals

General name “Product name”	Production base	Country
Fluoropolymers “Fluon® ” “Fluon+™ ”	■ Chiba plant ■ Kashima plant	Japan
	■ AGC Engineering Co., Ltd.	
	■ AGC Chemicals Europe, Ltd.	England
	■ AGC Chemicals Americas Inc.	Americas
Fluoroelastomers “AFLAS™”	■ Chiba plant	Japan
Fluoropolymer film “Fluon® ETFE FILM” Green house film “F-CLEAN™”		
Transparent Fluoropolymers “CYTOP”	■ Kashima plant	
Fluoropolymer for coatings “Lumiflon™”	■ Chiba plant	
Ion-exchange membranes “FORBLUE™ FLEMION™		
Fluorinated gases and solvents “AMOLEA™1234yf”, “AMOLEA™1224yd“		
Fluorinated solvent ”AE-3000”, “AC-6000” “AMOLEA™AS-300”	■ Chiba plant ■ Kashima plant	
Fluorocoating agents “SFCOAT™”	■ AGC Seimi Chemical	
Fine silica RESIFA™ M.S.GEL™, RESIFA™ SUNSPHERE™, RESIFA™ SUNLOVELY™	■ AGC Si-Tech	

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Life Science

Life Science Business Overview

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- Comprises Small molecule pharmaceuticals and Agrochemicals CDMO and Biopharmaceuticals CDMO

Small molecule pharmaceuticals and Agrochemicals CDMO

Providing high-quality CDMO services utilizing a wide range of chemical reactions, including fluorination, developed through our chemicals business.

Small molecule pharmaceuticals CDMO

Contract development and manufacturing of small molecule pharmaceuticals



Agrochemicals CDMO

Contract development and manufacturing of agrochemicals

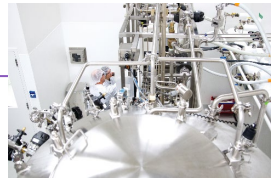


Biopharmaceuticals CDMO

Providing comprehensive CDMO services for biopharmaceuticals and gene/cell therapies, covering all stages from development to commercial production.

Contract development and manufacturing of biopharmaceuticals

- Mammalian
- Microbial



Contract development and manufacturing of gene and cell therapy

- Viral Vector
- Cell Therapy
- pDNA/mRNA
- Exosomes

CDMO: Contract Development and Manufacturing Organization

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Strengths of AGC's CDMO business

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Strengths

1. **Providing a high standard of services integrated across three regions of Japan, the US, and Europe**

2. **Flexible production system providing services from early-stage development to commercial phase**

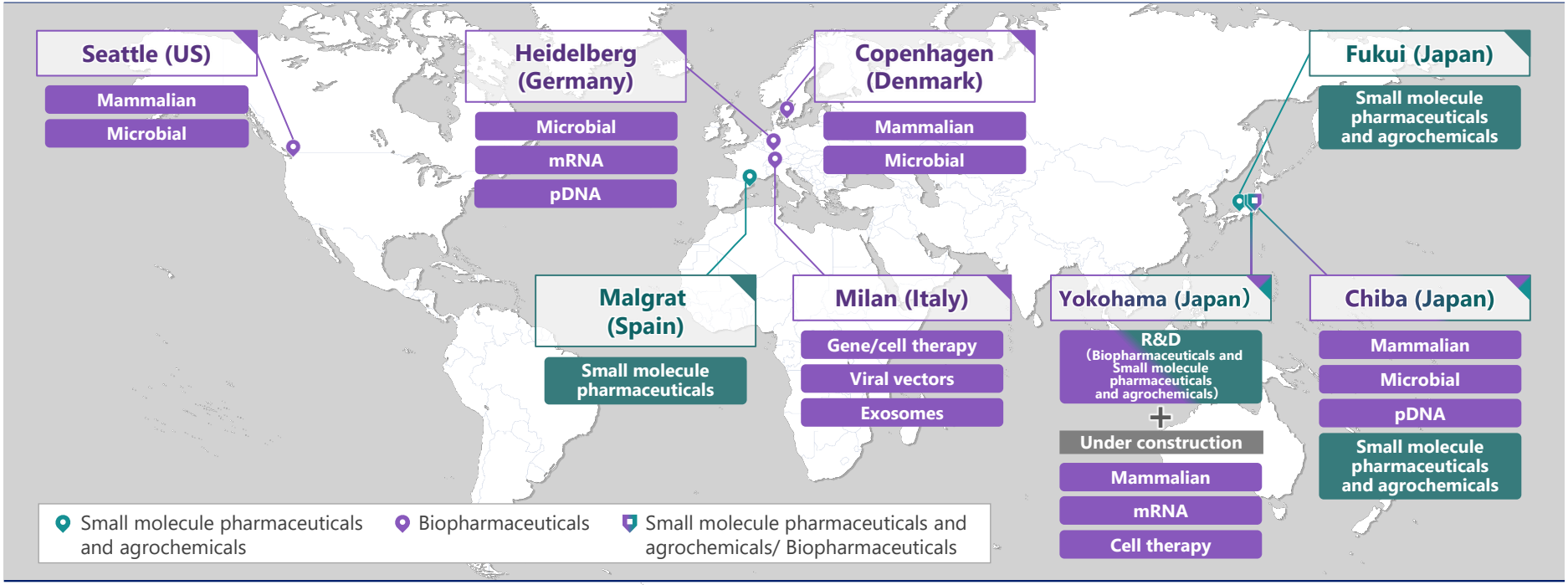
3. **Extensive Manufacturing and Inspection Track Record**

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Life Science Global Operations

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- We have established a highly integrated **cGMP system** in Japan, the U.S., and Europe, and provide the same **high standard of development and manufacturing services** in a **wide range of fields** from any of these regions.



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Strengths | Flexible to production needs from early-stage development to commercial scale

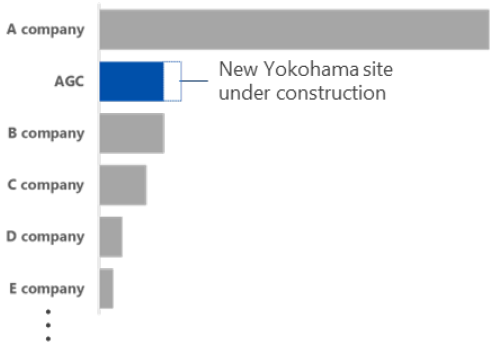
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- Addresses a wide range of production scale needs that vary in accordance with the progress of the development phase of the drug product

Small molecule pharmaceuticals	We have both pilot facilities suitable for small-volume production in the early stages of development and large reactors for commercial phase.
Biopharmaceuticals	AGC is a pioneer in the introduction of SUB*1, which enable flexible production of small- to medium-scale production. Current capacity ranks 2nd largest globally.

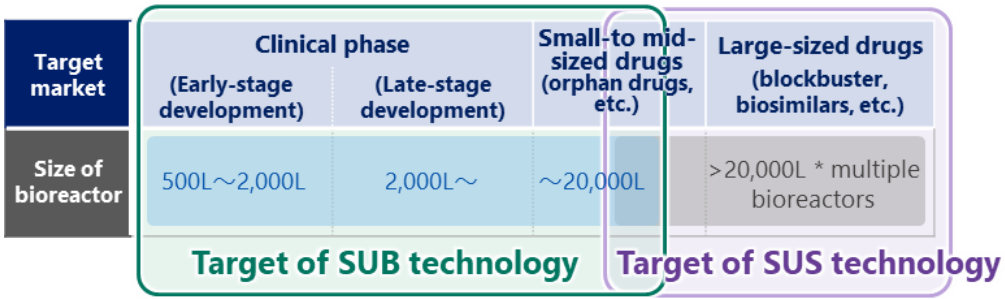
SUB Market Players

Antibody (mammalian cells): Production capacity of major competitors*2



Scope of AGC's services in the biopharmaceutical CDMO business

Combination of various bioreactor sizes to a **wide range of production scale and development phase.**



*1 SUB (Single-Use Bag) *2 As of June 2025 *3 SUS(stainless steel bioreactor)

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Strengths | Extensive Manufacturing and Inspection Track Record

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- With high-level quality and developmental capabilities, we have successfully undergone numerous authority inspections.

Inspection track record

		FDA U.S. Food and Drug Administration	EMA European Medicines Agency	PMDA Pharmaceuticals and Medical Devices Agency
Small Molecules	AGC Chiba Plant	●		●
	AGC Pharma Chemicals Europe Malgrat	●	●	●
Biopharmaceuticals	AGC Biologics Seattle	●	●	●
	AGC Biologics Copenhagen	●	●	●
	AGC Biologics Heidelberg	●	●	●
	AGC Biologics Milan	●	●	
	AGC Chiba Plant			●

* Including non-commercial products

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New Businesses : Solutions for next-generation high-speed communications

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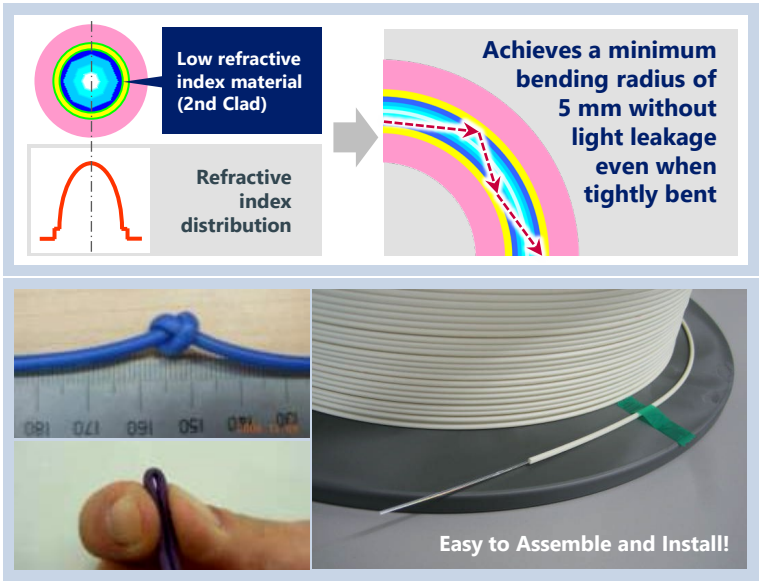
FONTEX®

FONTEX

A consumer fluorine-based plastic fiber featuring high transmission speed and ease of handling and connection

- FONTEX® is world's first optical fiber that enables high bandwidth and large capacity data communication with a transmission speed of 10 Gbps or more. The cable features resistance to tight bending, allowing data communication which could not have been achieved with existing silica-based fibers or POFs.
- Double-clad structure successfully prevents light from leaking out of the core. (See the figure in the upper left.) FONTEX® is an incredibly easy to handle optical fiber that doesn't have any kinks that inherently form in plastic and, above all, allows data communication under tight knots or bends. (See the figure in the lower left.)
- FONTEX® adopts a tight cord structure which the cable covering is directly attached to the fiber circumference (See the figure in the lower left.) This leads to cost reduction and simplified terminal treatment processing, allowing anyone to handle the cable with ease.
- FONTEX® is a safe and reliable plastic optical fiber (POF) for general household use. It is suitable for large-volume data transmissions, which are essential for next-generation 4K-TVs, and enables transmission even when it is handled roughly or is tightly bent.

Double-clad structure of FONTEX®

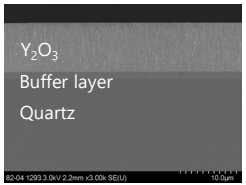


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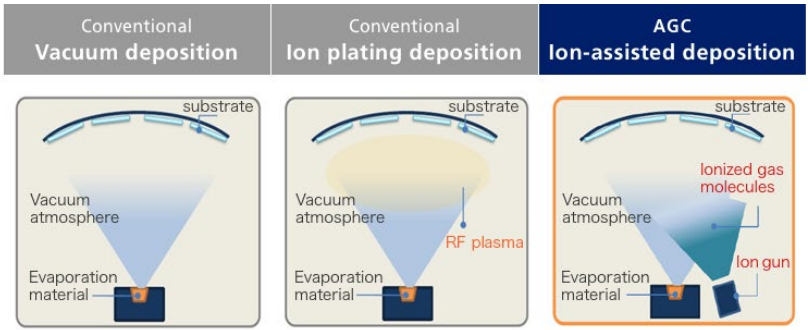
Yttrium Coating (Y2O3/Y5O4F7)

Ion-assisted vapor deposition method achieves dense, high hard coatings. Improved plasma resistance helps extend component life.

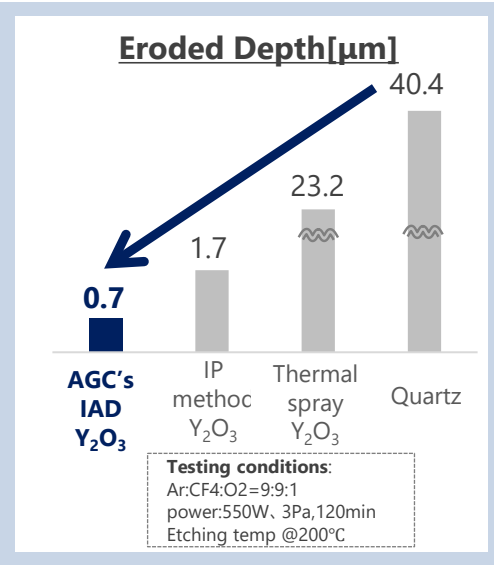
- The ion-assisted deposition (IAD) method achieves denser and harder coatings than those by vacuum evaporation and ion plating.
- The coatings have high corrosion resistance in environments requiring plasma resistance, extending the lifespan of equipment components and reducing maintenance costs.
- The capability to form dense coatings reduces particle generation, improving yields in manufacturing processes.







Method		Vacuum deposition	Ion plating deposition	Ion-assisted deposition
Substrate	Conductive substrate	△	○	◎
	Insulating substrate	△	△	◎



Ceramics / Others

Overview of AGC Ceramics products

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Common name	Name of typical product	Product description
Fused cast refractory	**ZB-X950 Series**	A high-performance refractory made by electro-fusing at an ultra-high temperature of over 1,800°C and casting. It is adopted as a key material for glass melting furnaces by many customers worldwide.
Monolithic refractory	**Asahi Caster®/GRAM**	Monolithic refractory is made by blending carefully selected refractory raw material through advanced mixing technology. It is used in various high-temperature plants, AGC Ceramics proposes solutions to meet customer needs such as "shorter installation period" and "energy saving".
Bonded refractory	**BNC/HAS**	A refractory material with both corrosion resistance and thermal shock resistance, made by press forming refractory raw materials and sintering them at a high temperature of 1,500°C or above. It is mainly used in cement kilns and has been adopted by many customers worldwide.
Advanced ceramics	**CERAROI-N/CERAROI-C**	Ceramics with excellent thermal and corrosion resistance made by forming submicron high-purity raw materials under high pressure and sintering them in a reduction atmosphere. They are used in many fields such as steel, aluminum, and electronics.
Ceramics material for 3D printers	**Brightorb**	A ceramics material optimized for ink jetting 3D printer that brings together the manufacturing know-how such as ceramics melting technology that we have accumulated over the years. It is attracting attention for shortening the lead time of the casting process and for craft and art applications.
Engineering	**Eco-lead furnace**	We contribute to energy saving, environmental load reduction, longer service life, and stable operation of glass furnaces by providing our engineering services to meet customer needs, including furnace design, construction support, and remote operation monitoring.

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Overview of the AGC Ceramics Group

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Japan

AGC Ceramics Co., Ltd.

- Headquarters: Tokyo
- Takasago Plant (Hyogo Prefecture) Manufacturing & Development
- Sales Offices:Tokyo, Osaka, and Kitakyushu

AGC Plibrico Co., Ltd.

Specialized manufacturer of monolithic refractories

- Headquarters: Tokyo
- Chigasaki Plant (Kanagawa Prefecture) Manufacturing & Development



(Tokyo) Mita NN Building

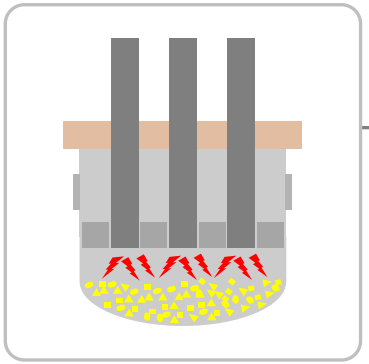


AGC Ceramics Takasago Plant office
(Within Takasago Factory of AGC Kansai Plant)

Overseas sites

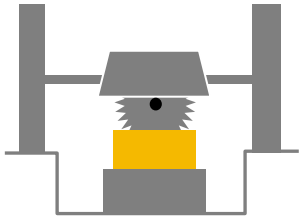
- **淄博艾杰旭剛玉材料有限公司**（China）
- **艾杰旭派力固（大連）工業有限公司**（China）
- **江蘇恒耐杰旭工业陶瓷有限公司**（China）
- **微瓷科技(江西)有限公司**（China）
- **AGC Ceramics Singapore Pte. Ltd.**

Fused cast refractory manufacturing process



- Preparing mold
- Electrical fusing
- Casting
- Annealing
- Processing
- Pre-Assembly Inspection
- Packing Shipment





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Fused cast refractory

- AGC offers a full line-up of fused cast refractories for glass melting furnaces to meet various customer needs.

ZrO_2 (zirconia) fused cast refractories ZB-X950 Series

Fused cast refractories with optimal performance against contamination to molten glass and excellent corrosion resistance required for use in glass melting furnace manufacturing high-quality specialty glass.

ZB-X950



Al_2O_3-ZrO_2-SiO_2 (AZS) fused cast refractories Zirconite Series

Fused cast refractories with excellent resistance to corrosion even under high temperatures

Zirconite



Al_2O_3 (alumina) fused cast refractories Marsnite Series

Fused cast refractories with excellent performance against contamination to molten glass

Marsnite



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Dry gunning Low cement monolithic refractory GRAM Series

To contribute to shortening installation period and reducing installation workers

- **Unique product that enables to install high-performance low cement monolithic refractory by dry gunning technique**
- Simple installation method and short heating-up time contribute to **shortening installation period**.
- It contributes to **reducing installation workers** as it can be installed without complicated wet gunning technique.

Image of GRAM construction method

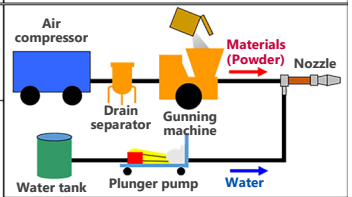
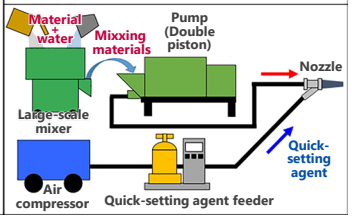
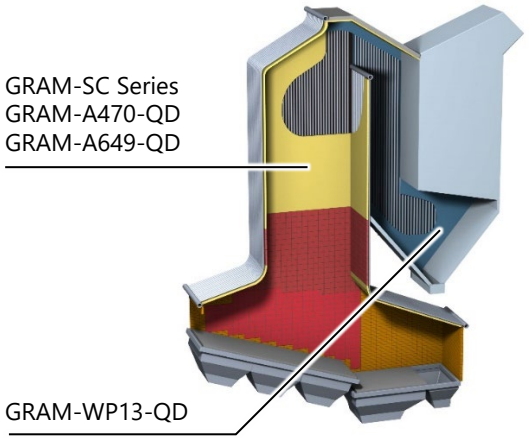
Construction method		Type of materials	Construction system
Dry gunning	Conven-tional	Cement-type (General-purpose)	
Dry gunning	GRAM	Low cement-type (High strength and high density)	
Wet gunning			

Image of GRAM lining

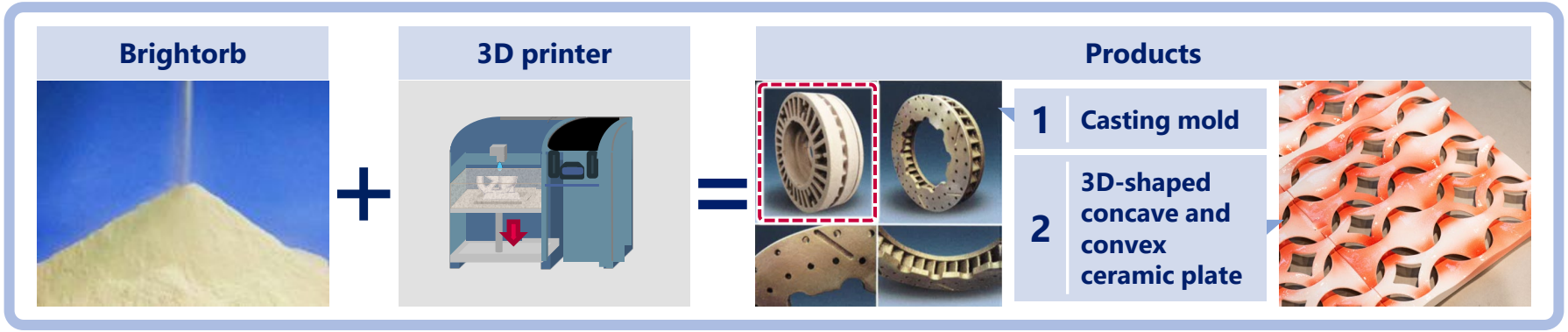


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Brightorb

Ceramics material for ink jetting 3D printers*

* AGC Ceramics has a large 3D printer with a 1,000 mm x 600 mm x 500 mm molding box.



Brightorb features

- Applicable to **molds** for cast metals of 1,600°C or higher such as cast-iron.
- **Shrinkage ratio is less than 1%**** after firing at 1,400°C

** Ordinary clay for earthenware has shrinkage of 10% or more and shows shrinkage and deformation after firing.

Value provided

1. **Designing of high-precision ceramics**
2. **Mass customized production**
3. **Shorter casting process for prototyping and development period**

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Reference Information

Successive Presidents

	Position	Name	Term	Memo
1	President	**Toshiya Iwasaki**	Sep. 1907 - Oct. 1930	
2	Chairman	**Sanjiro Yamada**	Mar. 1931 - Jun. 1939	President position is vacant.
3	President	**Masakichi Ono**	Jul. 1939 - Dec. 1943	
4	President	**Kamesaburo Ikeda**	Mar. 1943 - Mar. 1946	※
5	President	**Kanichi Morimoto**	Mar. 1946 - Dec. 1946	※
6	President	**Kikuo Mori**	Dec. 1946 - Apr. 1950	※
7	President	**Tokiichiro Kuwata**	Apr. 1950 - Jun. 1950	※
8	President	**Kiichi Watanabe**	Jun. 1950 - Feb. 1952	
9	President	**Kanichi Morimoto**	Feb. 1952 - Aug. 1967	
10	President	**Motoharu Kurata**	Aug. 1967 - Feb. 1973	
11	President	**Hideaki Yamashita**	Feb. 1973 - Mar. 1981	
12	President	**Takeo Sakabe**	Mar. 1981 - Jan. 1987	
13	President	**Jiro Furumoto**	Jan. 1987 - Mar. 1992	
14	President	**Hiromichi Seya**	Mar. 1992 - Jun. 1998	
15	President	**Shinya Ishizu**	Jun. 1998 - Mar. 2004	
16	President	**Masahiro Kadomatsu**	Mar. 2004 - Mar. 2008	
17	President	**Kazuhiko Ishimura**	Mar. 2008 - Dec. 2014	
18	President	**Takuya Shimamura**	Jan. 2015 - Dec. 2020	
19	President	**Yoshinori Hirai**	Jan. 2021 -	

※ April 1944 ～ May 1950 , （Former）Mitsubishi Kasei Kogyo Co., Ltd

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Corporate History (1)

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1907 – Asahi Glass Company is established.

1909 – Flat glass production begins at Amagasaki (now Kansai) for the first time in Japan

1914 – Makiyama (now Kitakyushu) Plant is established.

1916 – Production of refractories begins and ceramics business launches.
– Tsurumi Plant is established.

1917 – Production of soda ash begins.

1939 – Iho (now Takasago) Plant is established.

1944 – Merges with Nippon Chemical Industries and Mitsubishi Chemical Industries is formed.

1950 – Pursuant to the Company Reconstruction and Improvement Law, Mitsubishi Chemical Industries is divided into Asahi Glass, Nippon Kasei Kogyo (now Mitsubishi Chemical) and Shinko Rayon (now Mitsubishi Rayon). Asahi Glass Company is re-established and lists its stock on the stock exchanges in Japan.

1954 – Production of CRT glass bulbs begins.

1956 – Production of CRT glass bulbs begins.
– The Indo Asahi Glass Co., Ltd. is established, and business launches in India.

1959 – Chiba Plant is established.

1964 – Caustic soda first is produced at Chiba Plant.
– Thai Asahi Glass Public Co., Ltd. is established and business launches in Thailand.

1965 – Hazawa Research Center (now Research Center) is established.
– Thai Asahi Caustic Soda Co. (now AGC Chemicals (Thailand) Co., Ltd.) is established and production of chemicals products begins in Asia.

1970 – Aichi Plant is established.

1972 – Sagami Works (now Sagami Plant) is established.
– P.T. Asahimas Flat Glass Co., Ltd. is established and business launches in Indonesia.

1974 – Kashima Plant is established
– Thai Safety Glass Co., Ltd. (now AGC Automotive (Thailand) Co., Ltd.) is established and automotive glass production begins in Asia.

1981 – Asahi Glass acquires Glaverbel S.A., (AGC Glass Europe) a glass company in Belgium, and launches business in Europe.

1985 – AP Techno glass Co. is established automotive glass production begins in the USA.
– Production of synthetic quartz glass begins.

1986 – P.T. Asahimas SUBENTRA Chemical (now P.T. Asahimas Chemical) established in Indonesia.

1988 – Makes capital participation into AFG Industries (now AGC Flat Glass North America) and flat glass production begins in the USA.

1991 – Makes capital participation into Splintex S.A. (now AGC Automotive Europe Inc) and automotive glass production begins in Europe.
– Makes capital participation into Glavunion (now AGC Flat Glass Czech a.s. clen AGC Group) and launches business in Czech Republic.

1992 – Dalian Asahi Float Glass Co., Ltd. (now AGC Flat Glass (Dalian) Co., Ltd.) is established and flat glass production begins in China.
– Production of TCO glass substrates for LCDs begins at Asahi Glass Fine Techno (now AGC Display Glass Yonezawa Co., Ltd.).

1995 – Production of alkali-free glass for TFT-LCD glass substrates begins.
– Qinhuangdao Haiyan Safety Glass Co., Ltd. is established and production of automotive glass business begins in China.

1996 – Production of glass substrates for Plasma Display Panels (PDPs) begins.

1997 – Makes capital participation into Bor Glass Works (now OJSC AGC Bor Glassworks) and launches business in Russia.

1999 – Acquires Imperial Chemical Industries PLC's fluoropolymer business and launches production of fluorochemicals in Europe.

2000 – Asahi Glass Fine Techno Taiwan Co., Ltd. (now AGC Display Glass Taiwan Co., Ltd.) is established and production of TFT-LCD glass substrates begins.

2002 – Global In-House Company System is introduced.

2003 – Production of PDP glass substrates begins at Hanwook Techno Glass Co., Ltd.

Corporate History (2)

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2004
- AGC Automotive Hungary Ltd. is established and automotive glass production begins in Hungary.
- Asahi Glass Fine Techno Korea Co., Ltd. (now AGC Fine Techno Korea Co., Ltd.) is established and production of TFT-LCD glass substrates begins in South Korea.

2007
- The corporate brand of the AGC Group is unified to AGC.
- Transfers the shares of Asahi Fiberglass Co., Ltd. and withdraws from glass fiber business.

2008
- Transfers the shares of Optrex Corporation and withdraws from LCD display device business.

2009
- Terminates automotive glass business at Kitakyushu plant.
- Begins production of specialty glass for chemical strengthening for smart phones, tablet PCs, etc.

2010
- AGC Display Glass (Kunshan) Co., Ltd. s established as a production base for TFT-LCD glass substrates in China.
- Terminates production of CRT glass bulbs at Hankuk Electric Glass Co., Ltd. and withdraws from CRT glass bulb business.

2011
- AGC Glass Brazil inc. is established and business launches in Brazil.

2012
- Strategic partnership with Interpane in Germany.

2013
- AGC Automotive Mexico S. DE R.L. DE C.V. is established.
- Established regional headquarters, AGC Asia Pacific, in Singapore for business in Southeast Asia.

2014
- Acquires a Vietnamese polyvinyl chloride (PVC) company, Phu My Plastics & Chemicals Co., Ltd. .(now AGC Chemicals Vietnam Co., Ltd.)
- AGC and Obeikan Glass agreed to set up a JV for architectural glass.
- Ceased production of glass substrates for PDP.

2015
- Acquires NordGlass, automotive replacement glass company in Poland.
- Started supplying HFO-1234yf, a next-generation automotive refrigerant with low environmental impact.

2016
- Acquired German biopharmaceutical contract manu-facturing organization BIOMEVA.

2017
- Acquires Vinythai PCL (shareholding ratio of 58.77%)
- Bioscience businesses in Japan, the U.S., and Europe consolidated and integrated management as AGC Biologics begins.

2018
- Change of corporate name to AGC Inc. (1st July)
- AGC acquires the electronics business of US-based Park Electrochemical.

2019
- AGC's First Automotive Glass Production Base is established in North A frica (Morocco).
- Acquires Spanish Synthetic Pharmaceutical Active Ingredient Manufacturing Plant Malgrat Pharma Chemicals
- Acquires US based Taconic's global operations of the Advanced Dielectric Division

2020
- AGC Acquires Biopharmaceutical Commercial Facility in Colorado, U.S.A.
- AGC Completes Development of 5G-compatible 'Glass Antenna that Adds Cellular Base Station Capabilities to Windows'
- AGC to Make Drastic Expansion to Supply System for EUVL Mask Blanks.
- AGC Biologics Expands Cell and Gene Therapy CDMO Services.
- AGC Opens New R&D Building.

2021
- Integrated the research and development bases of the former Central Research Laboratory and the former Keihin Plant and began operations as the AGC Yokohama Technical Center.
- North American architectural glass business transferred to Cardinal Glass Industries.

2022
- Established a new company, AGC Vinythai Public Company Limited to integrate and reorganize the chlor-alkali business subsidiaries in Thailand and CLMV markets.

2023
- Established a new in-house company, Life Science Company.
  Completed the transfer of its ownership stake in AGC Flat Glass (Dalian) Inc. to Shanghai Yaohua Pilkington Glass Group Co. Ltd. (SYP Group)

2024
- Divestment of architectural and automotive glass businesses in Russia; withdrawal from the country.

AGC's Overseas Business Development

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Europe/EMEA

1981 – **【Belgium】** Asahi Glass acquires Glaverbel S.A., (AGC Glass Europe S.A.) a glass company in Belgium. First entry into Europe.

1997 – **【Russia】** Capital participation made in Bor Glassworks (OJSC AGC Bor Glassworks). First entry into the Russian market.

1999 – **【UK】** Imperial Chemical Industries PLC's (AGC Chemicals Europe, Ltd.) fluoropolymers business acquired. Production of fluorochemicals begins in Europe.

2004 – **【Hungary】** AGC Automotive Hungary Ltd. established. Production of automotive glass begins in Hungary.

– **【Saudi Arabia】** AGC and Obeikan Glass agreed to set up a JV for architectural glass.

2012 – **【Germany】** Strategic partnership with Interpane in Germany. (Became a consolidated subsidiary)

2015 – **【Poland】** AGC acquires NordGlass, automotive replacement glass company in Poland.

2019 – **【Morocco】** Automotive Glass Production Base established in Morocco.

– **【Spain】**Acquires Spanish Synthetic Pharmaceutical Active Ingredient Manufacturing Plant Malgrat Pharma Chemicals.

2020 – **【Italy】** AGC Biologics Expands Cell and Gene Therapy CDMO Services.

ASIA/Japan

1907 – **【Japan】 1907 Asahi Glass Company founded.**

1956 – **【India】** Glass production begins in India. AGC becomes one of the first Japanese private companies to enter the Indian market.

1964 – **【Thailand】** Thai Asahi Glass Public Co., Ltd. (AGC Flat Glass (Thailand) Public Co., Ltd.) established.

1972 – **【Indonesia】** P.T. Asahimas Flat Glass Co., Ltd. established in Indonesia. First entry into the Indonesian market.

1992 – **【China】** Dalian Asahi Float Glass Co., Ltd. established in China. Production of flat glass begins in China.

2000 – **【Taiwan】** Asahi Glass Fine Techno Taiwan Co., Ltd. (AGC Display Glass Taiwan Co., Ltd.) established. Production of LCD glass substrates begins in Taiwan.

2003 – **【Korea】** At Hanwook Techno Glass Co., Ltd., production of PDP glass substrates begins in Korea.

2014 – **【Vietnam】** AGC acquire a Vietnamese polyvinyl chloride (PVC) company, Phu My Plastics & Chemicals Co., Ltd.

2017 – **【Thailand】** Acquires Vinythai PCL.

2022 – **【Thailand】** established a new company, AGC Vinythai Public Company Limited to integrate and reorganize the chlor-alkali business subsidiaries in Thailand and CLMV markets.

Americas

1985 – **【USA】** AP Technoglass Co. (Automotive glass division in AGC Flat Glass North America, Inc.) established. Production of automotive glass begins in the USA.

2011 – **【Brazil】** AGC Glass Brazil, Inc. established. First entry into the Brazilian market.

2013 – **【Mexico】** AGC Automotive Mexico S. DE R.L. DE C.V. established.

2018 – **【USA】** AGC acquires the electronics business of US-based Park Electrochemical.

– **【USA】** AGC Biologics, consolidated and integrated management for bioscience business.

2019 – **【USA】** AGC acquires global operations of the Advanced Dielectric Division of US based Taconic.

2020 – **【USA】** AGC Acquires Biopharmaceutical Commercial Facility in Colorado, U.S.A.

*This list shows AGC's first entry into the region.

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END

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